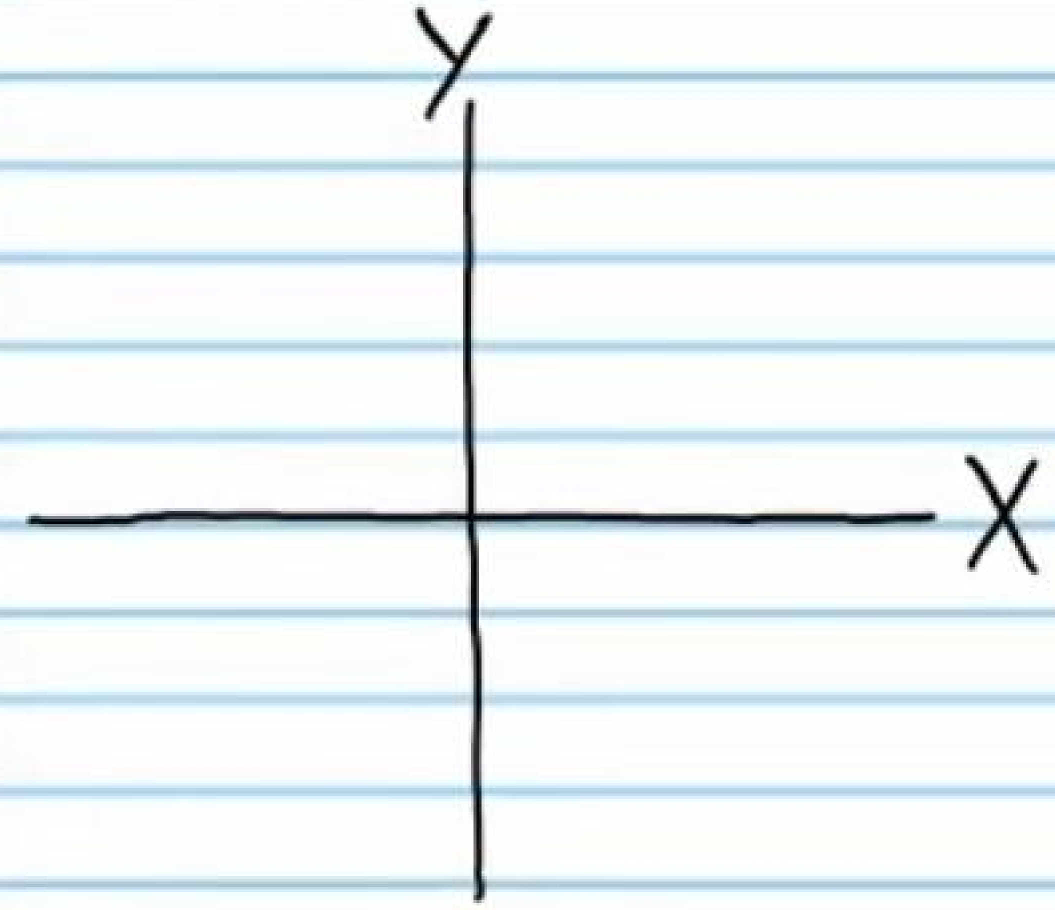
Name
Date
Course

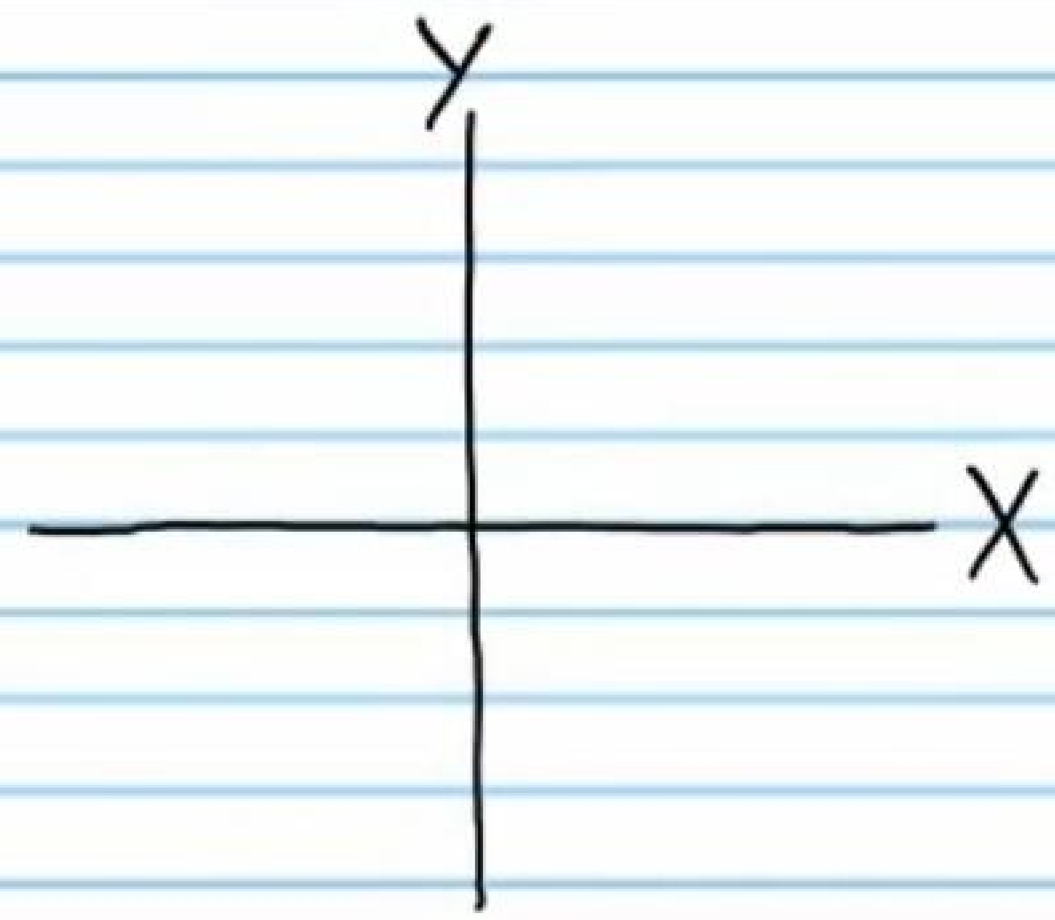
Continued Exploration of the Six Basic Trigonometric Functions (Homework)

Draw each angle in standard position, write the measure of the reference angle, and draw a triangle to help find the value of the trigonometric expression. Then, find the value of the trigonometric expression without a calculator.

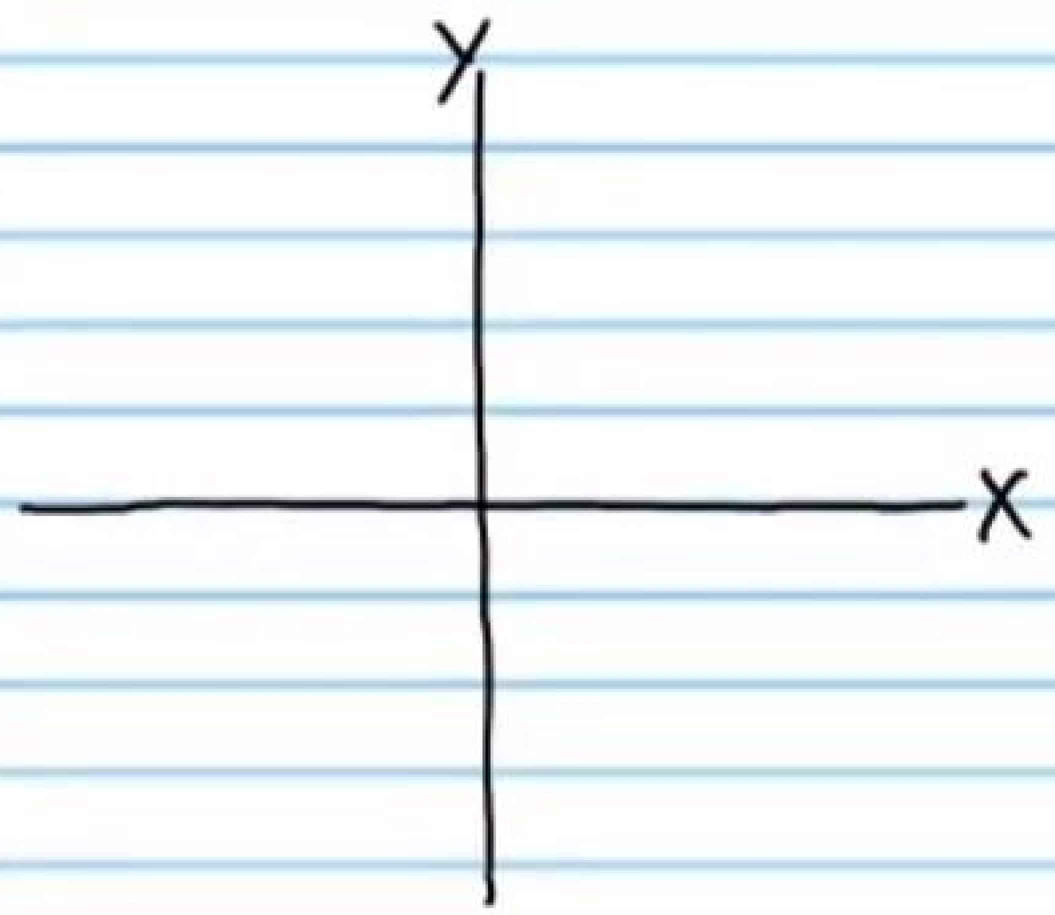
① $\cos(-45°)$



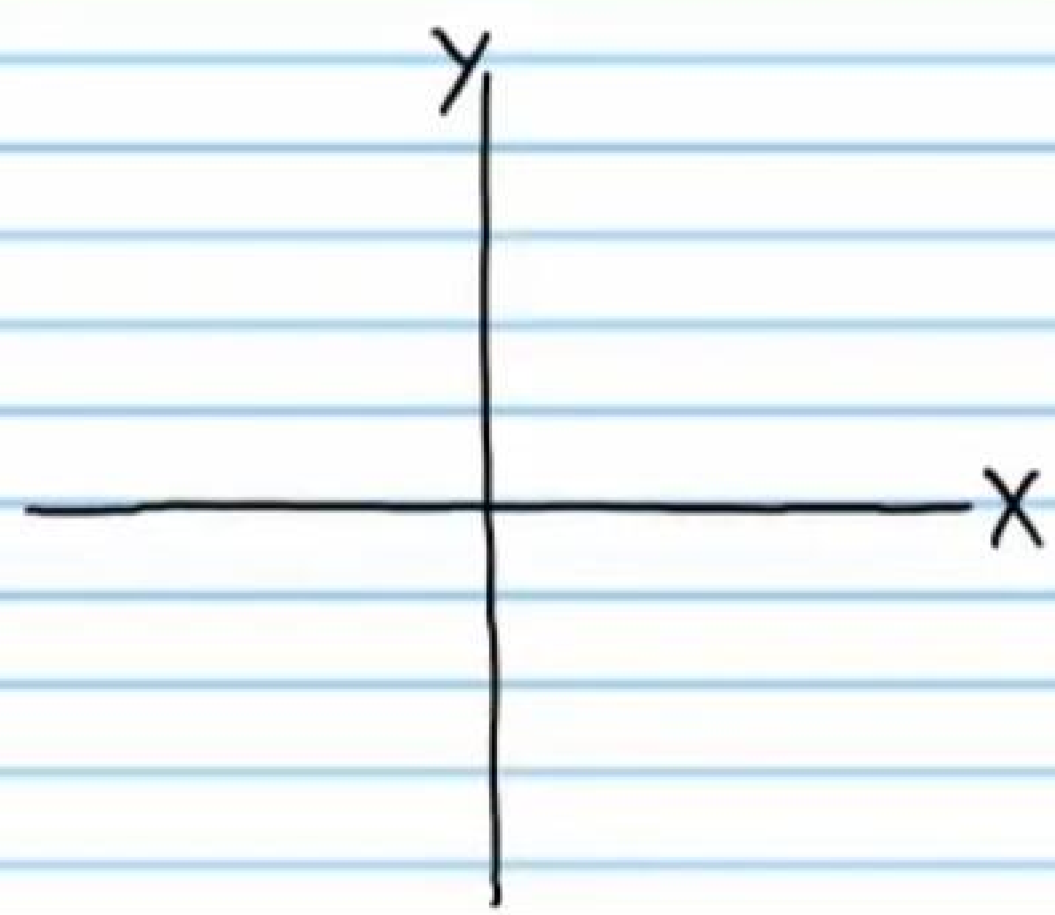
② $\cot(-210°)$



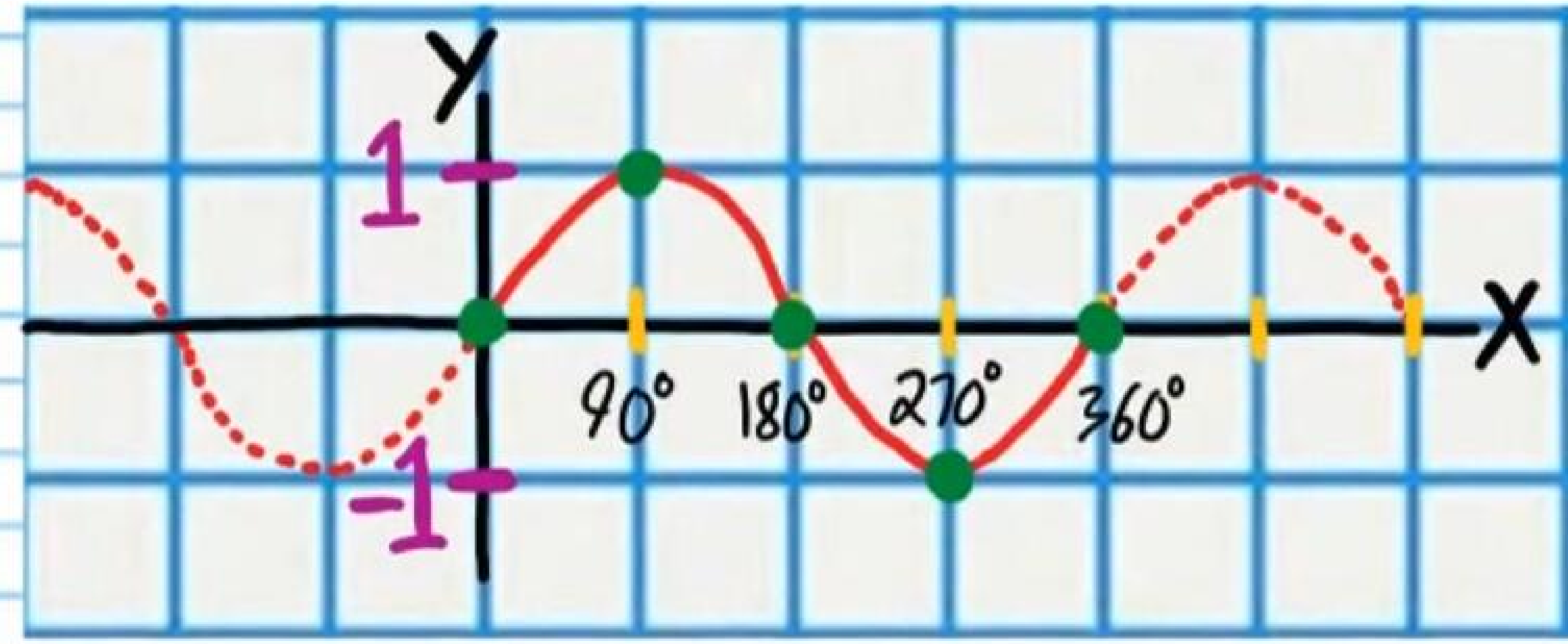
③ $\sin(-315°)$



④ $\sec(-120°)$



$y = \sin x$

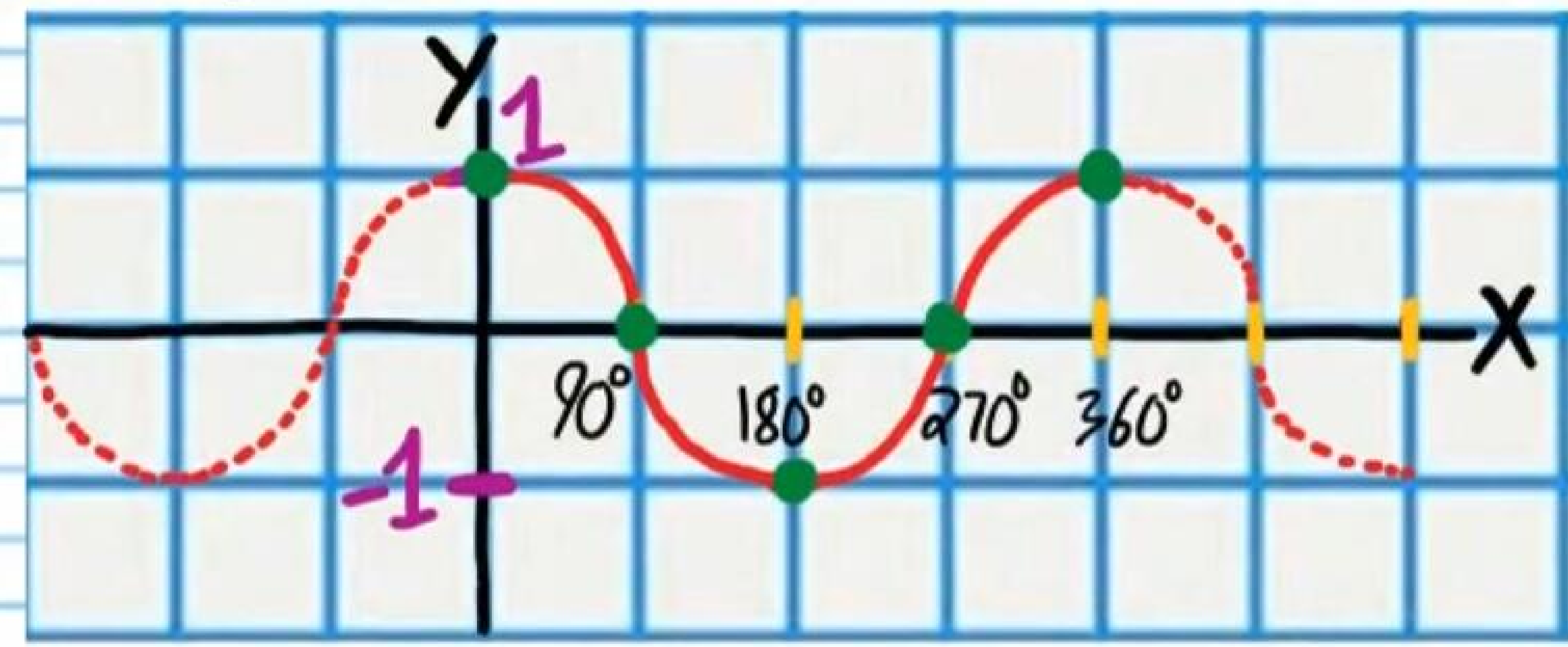


Use the graph above to find the following values.

⑤ $\sin 90°$ ⑥ $\sin 180°$ ⑦ $\sin 540°$

⑧ $\sin(-270°)$

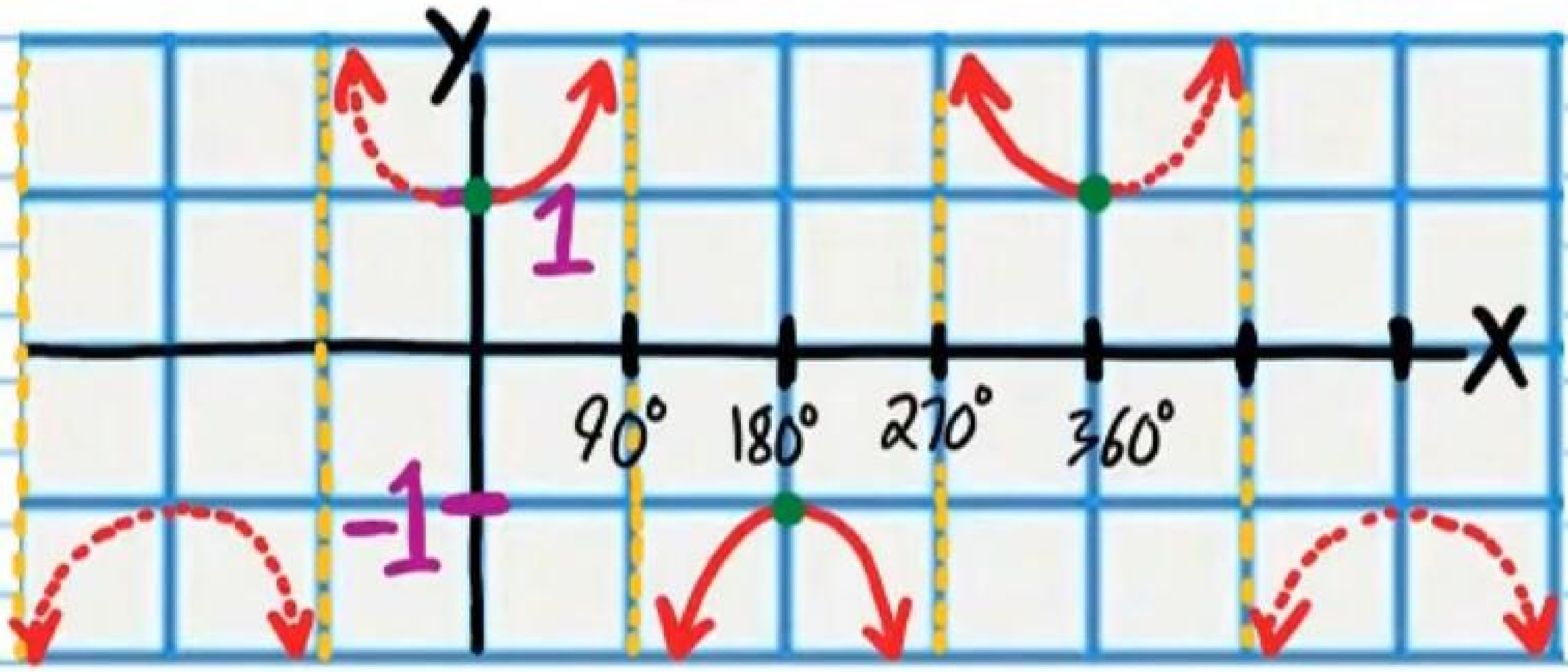
$y = \cos x$



Use the graph above to find the following values.

⑨ $\cos 450°$ ⑩ $\cos(-180°)$ ⑪ $\cos(-270°)$

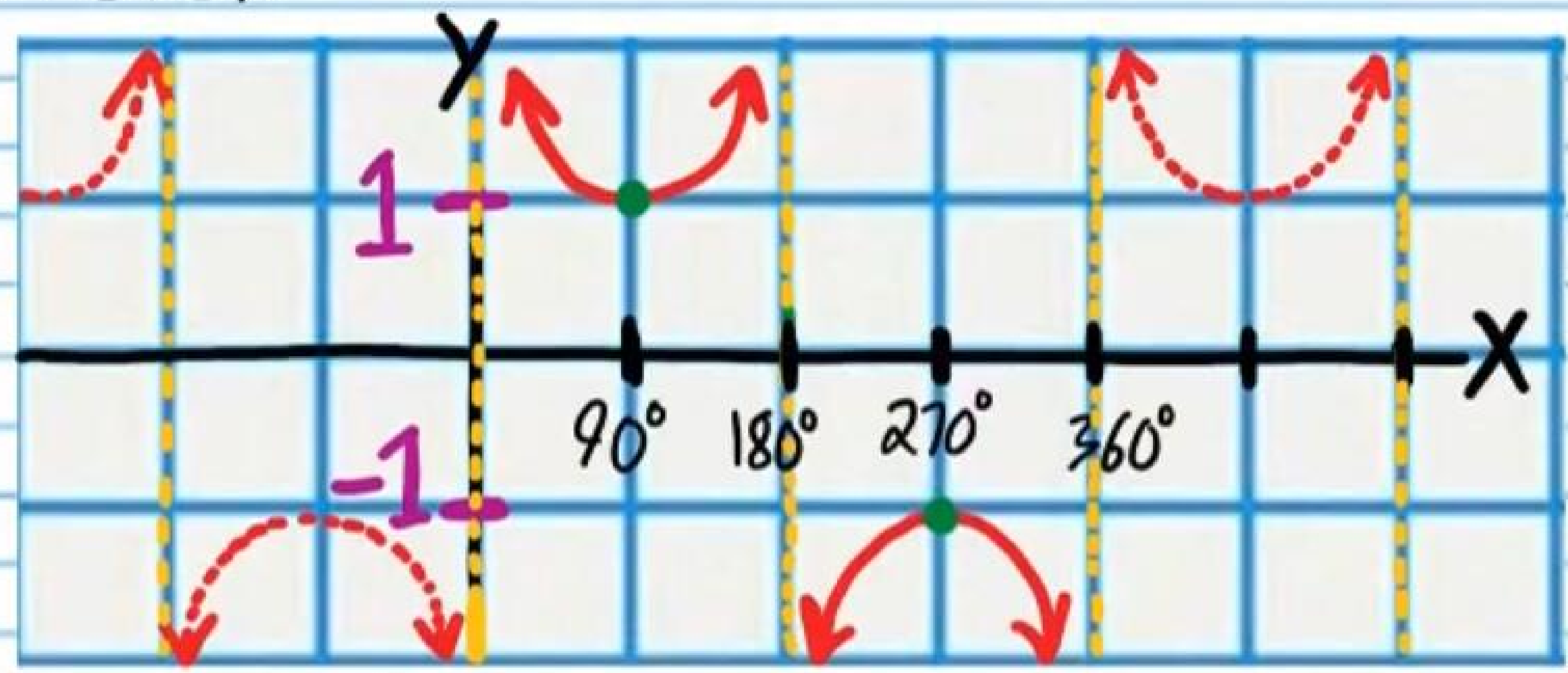
$y = \sec x$



Use the graph above to find the following values.

⑫ $\sec(-270°)$ ⑬ $\sec 360°$

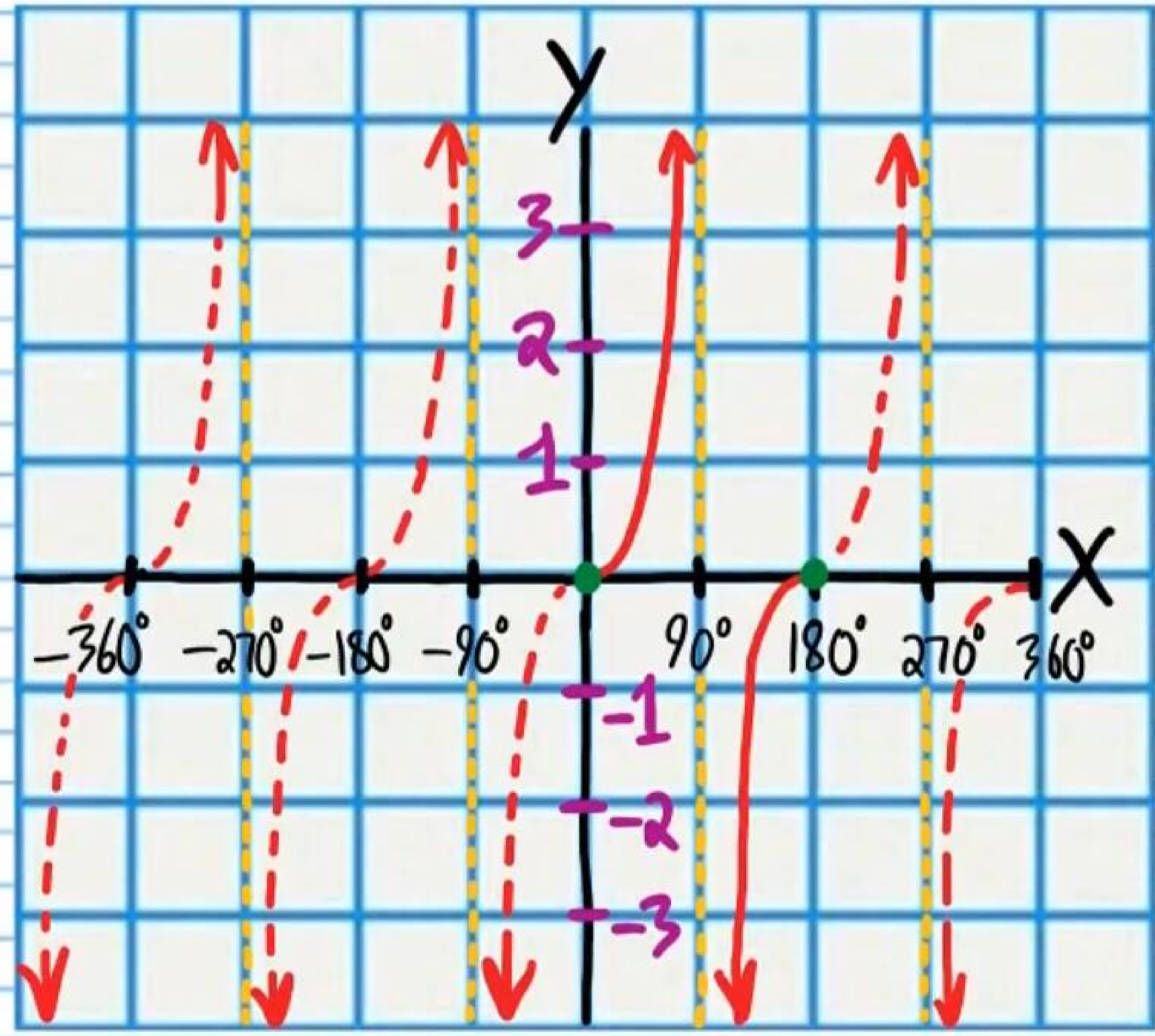
$y = \csc x$



Use the graph above to find the following values.

⑭ $\csc(-180°)$ ⑮ $\csc 450°$

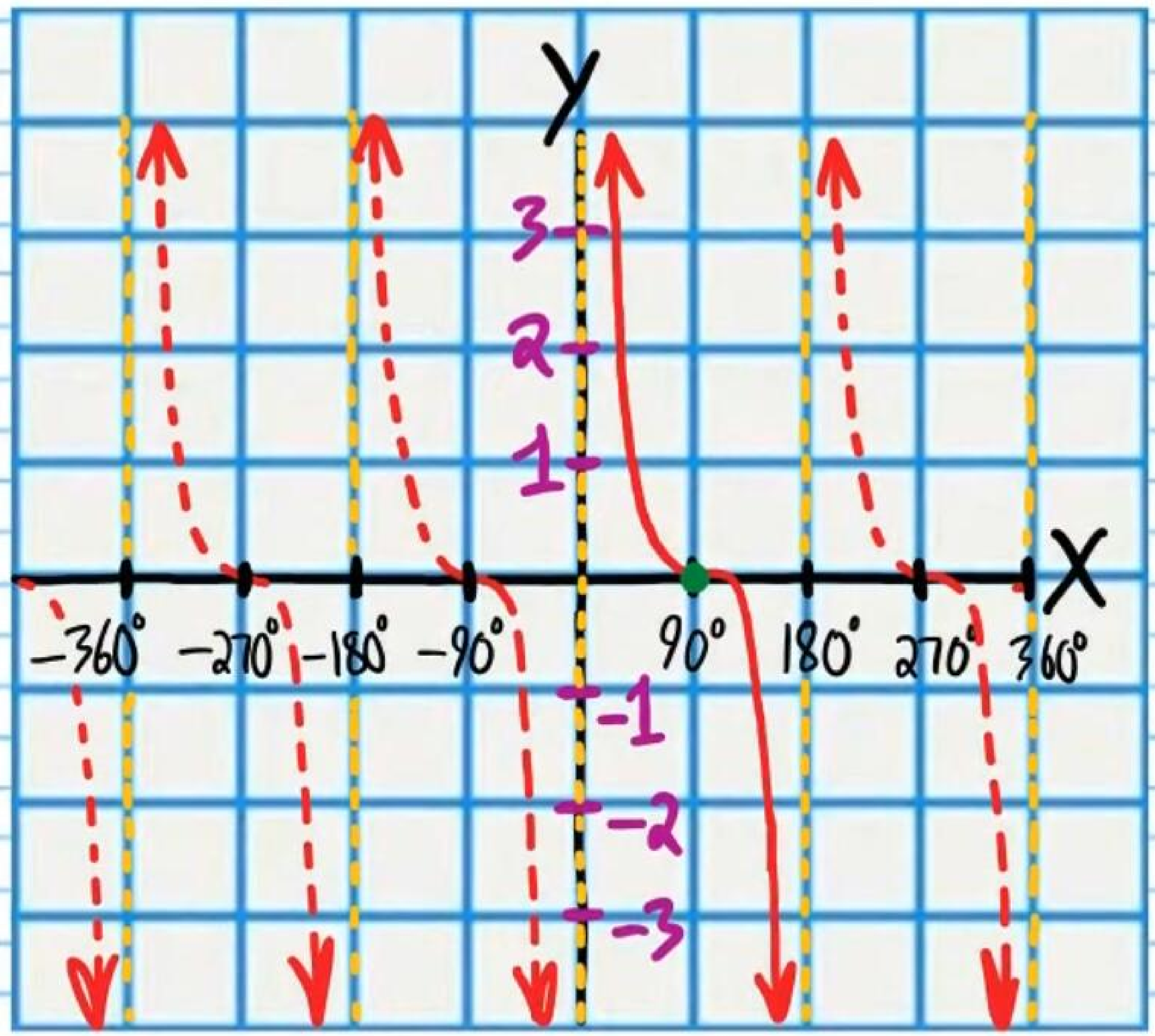
$y = \tan x$



Use the graph above to find the following values.

⑯ $\tan(-90°)$ ⑰ $\tan 270°$

$y = \cot x$



Use the graph above to find the following values.

(18) $\cot(-180°)$

(19) $\cot(-270°)$

The measures of angles in standard position are written below. Find the measures of three coterminal angles in standard position.

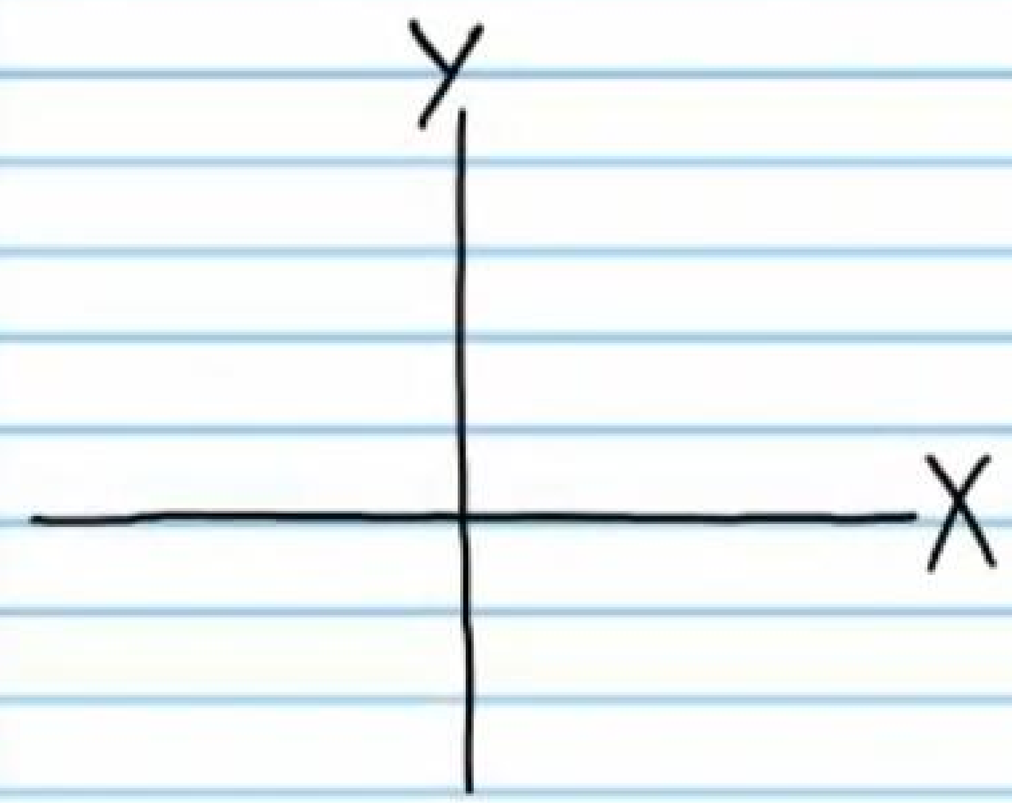
(20) $170°$

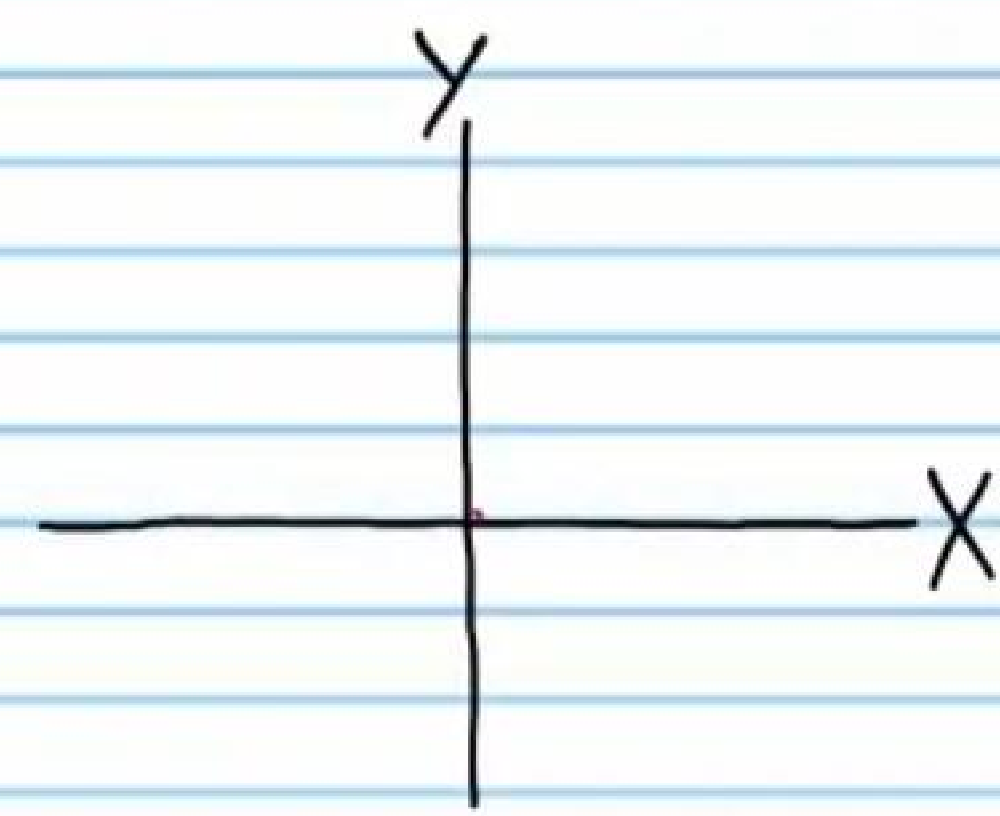
(21) $-42°$

(22) $311°$

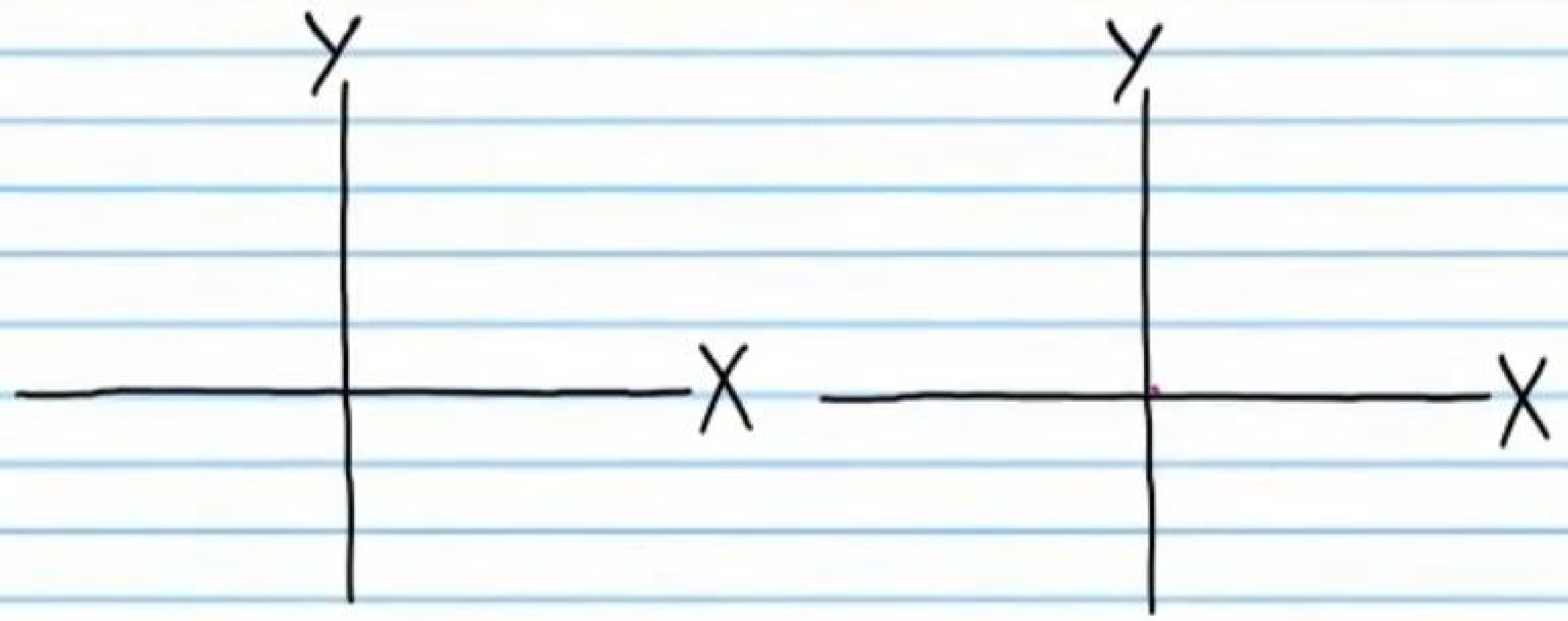
Use a calculator to show that the following equations are true. Then, draw a reference triangle in standard position to show why each equation is true.

(23) $\sin 130° = \sin 490°$

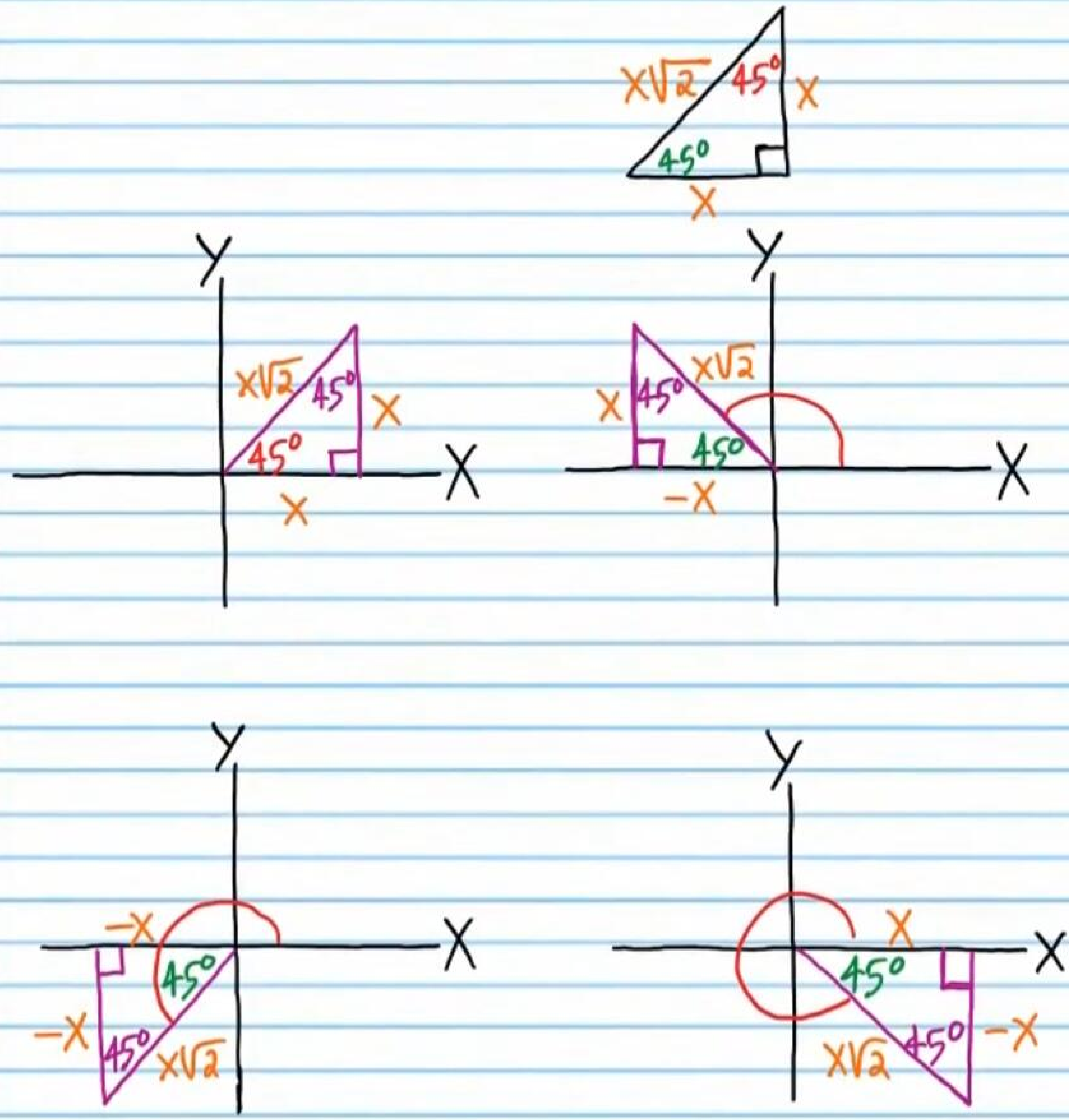




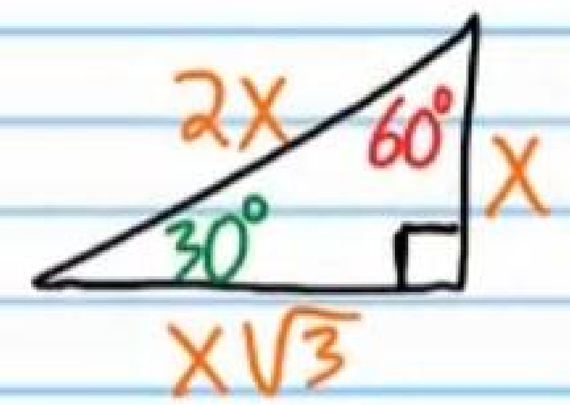
(24) $\tan(-40°) = \tan 320°$

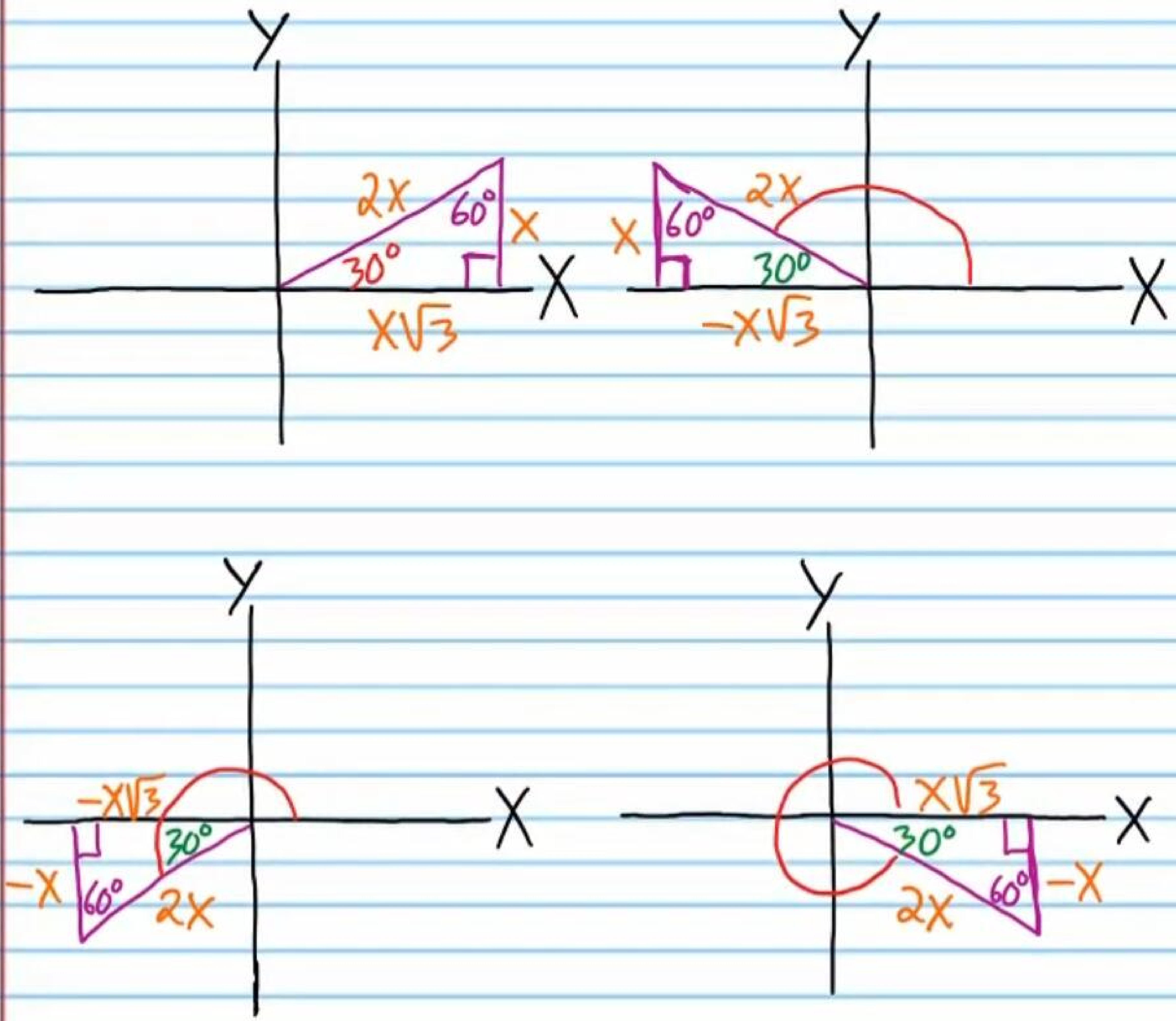


(25) If $\sin\theta = \frac{1}{\sqrt{2}}$, and $0° < \theta \le 360°$, Find the values of θ.

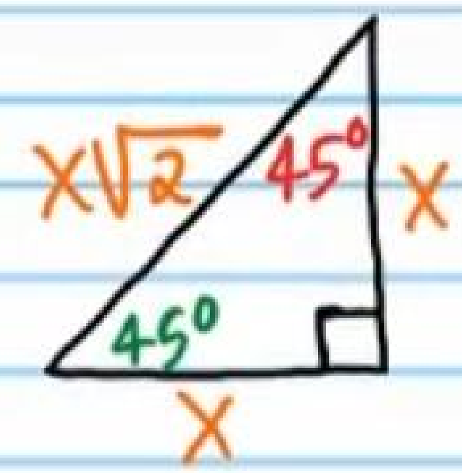


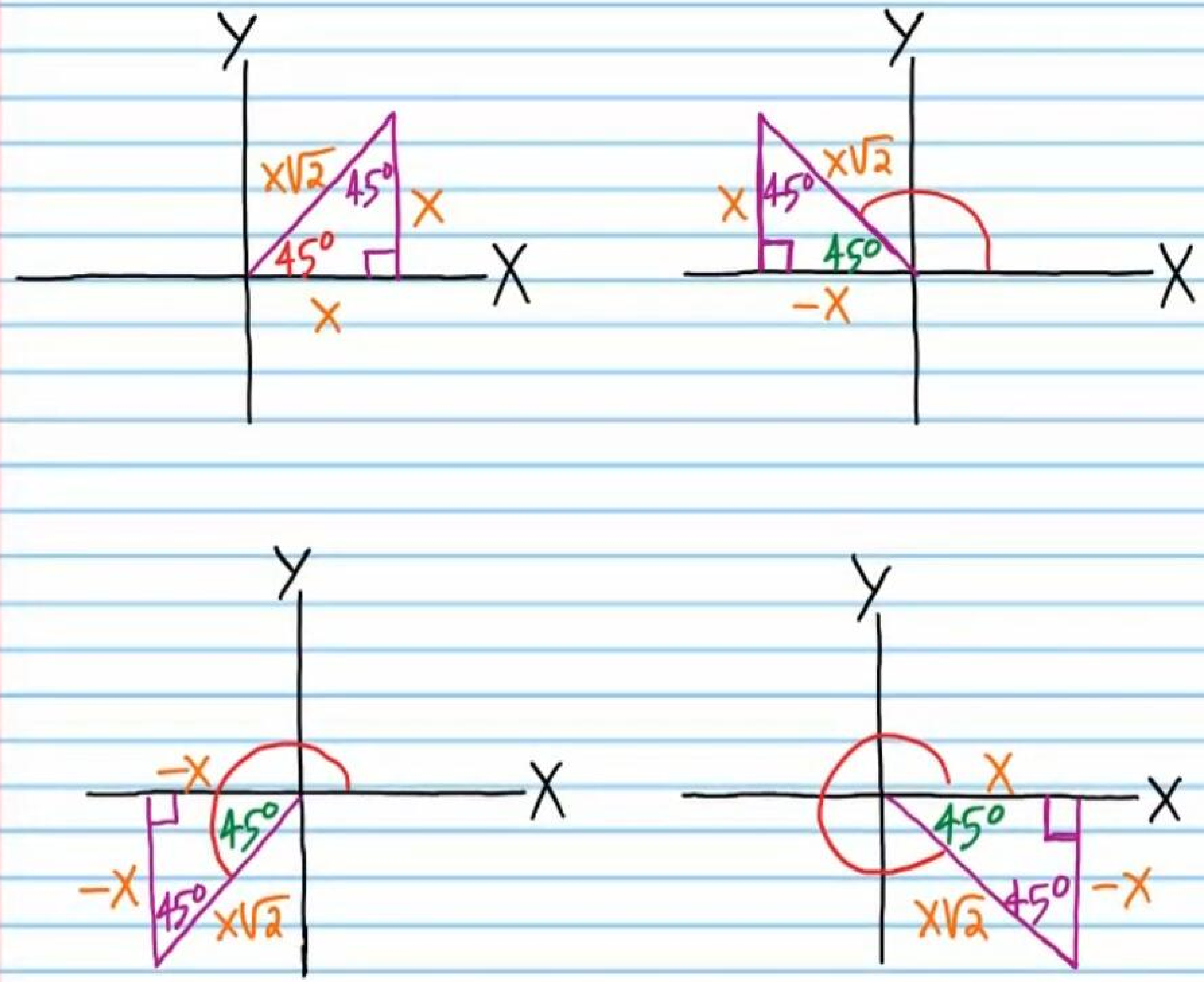
(26) If $\sec\theta = -\frac{2}{\sqrt{3}}$, and $0^\circ < \theta \le 360^\circ$, Find the Values of θ.



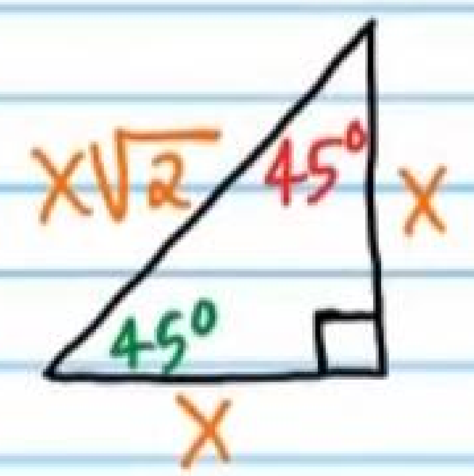


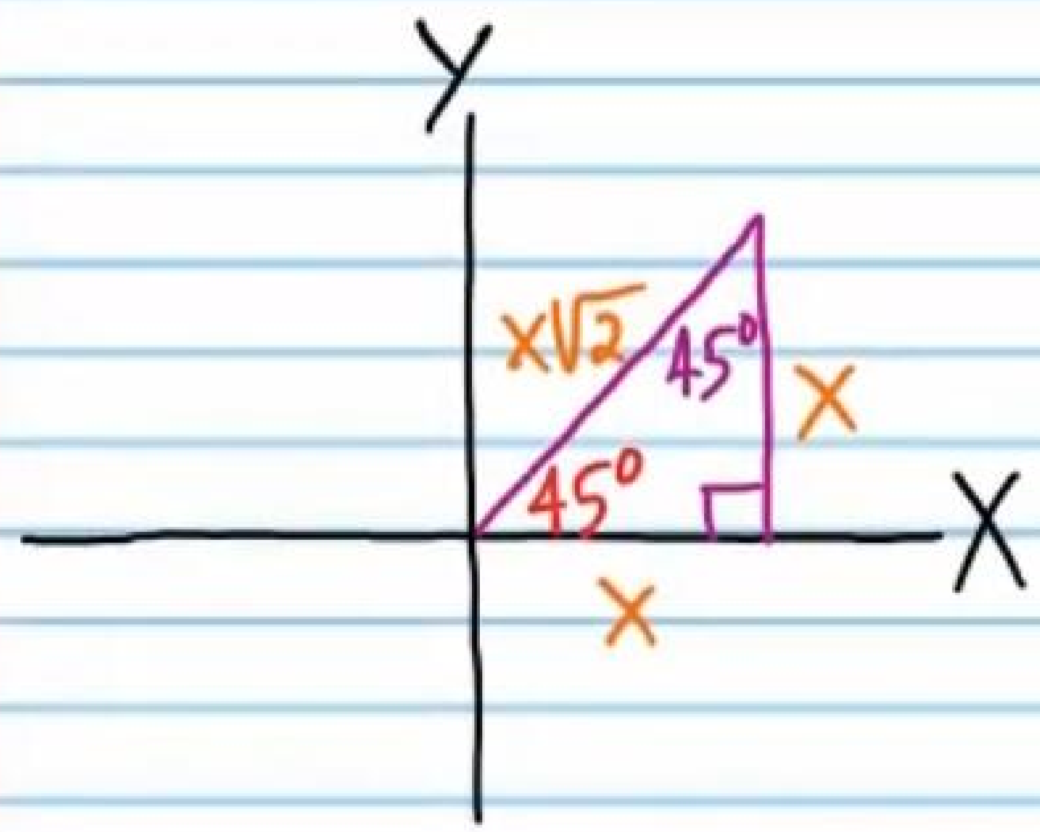
(27) If $\csc\theta = -\sqrt{2}$, and $0^\circ < \theta \le 360^\circ$, Find the Values of θ.

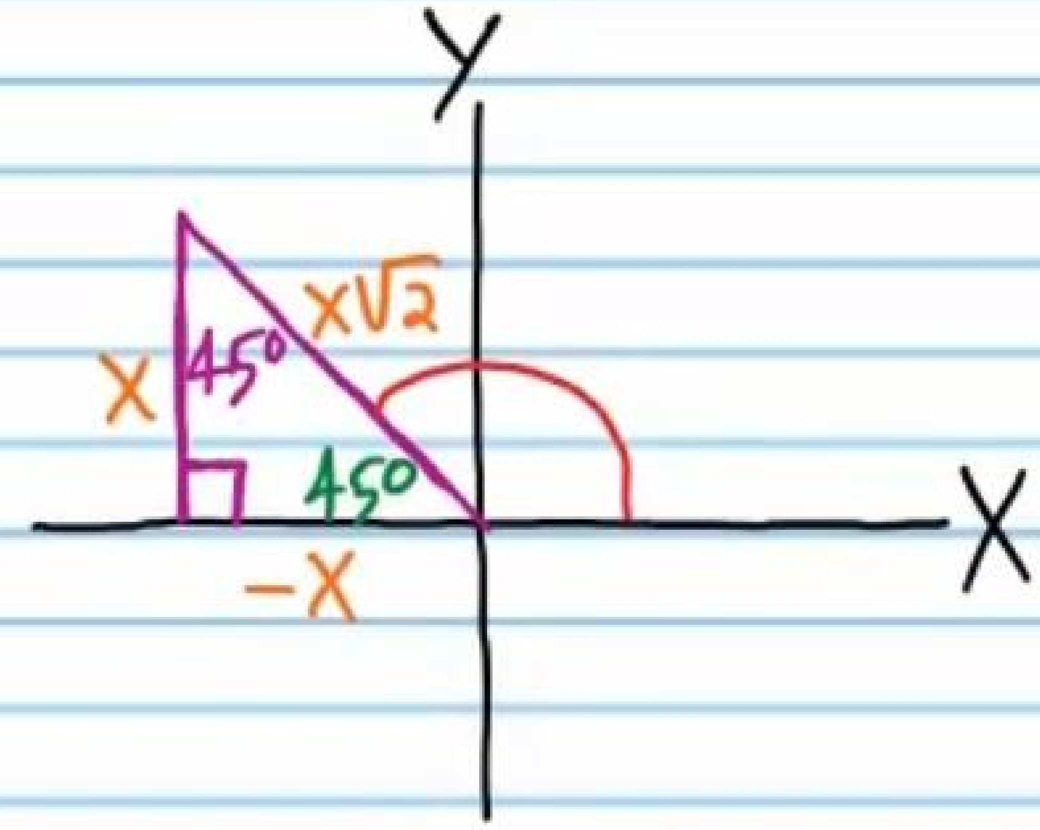


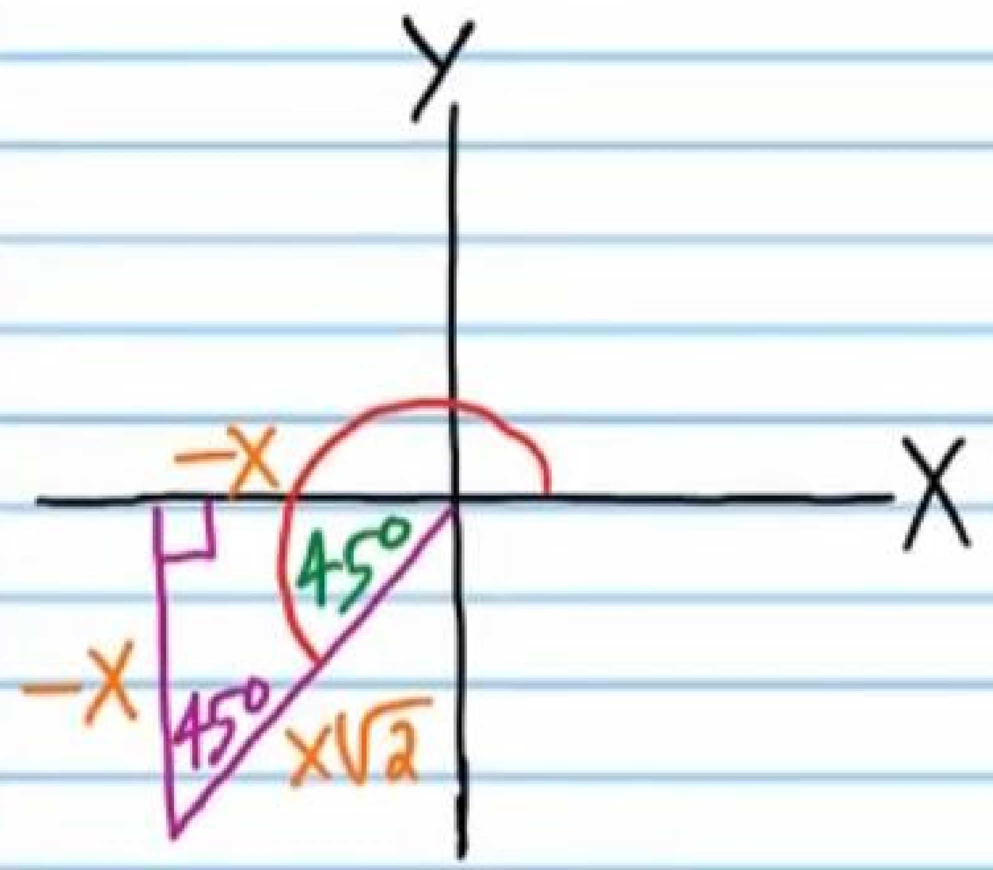


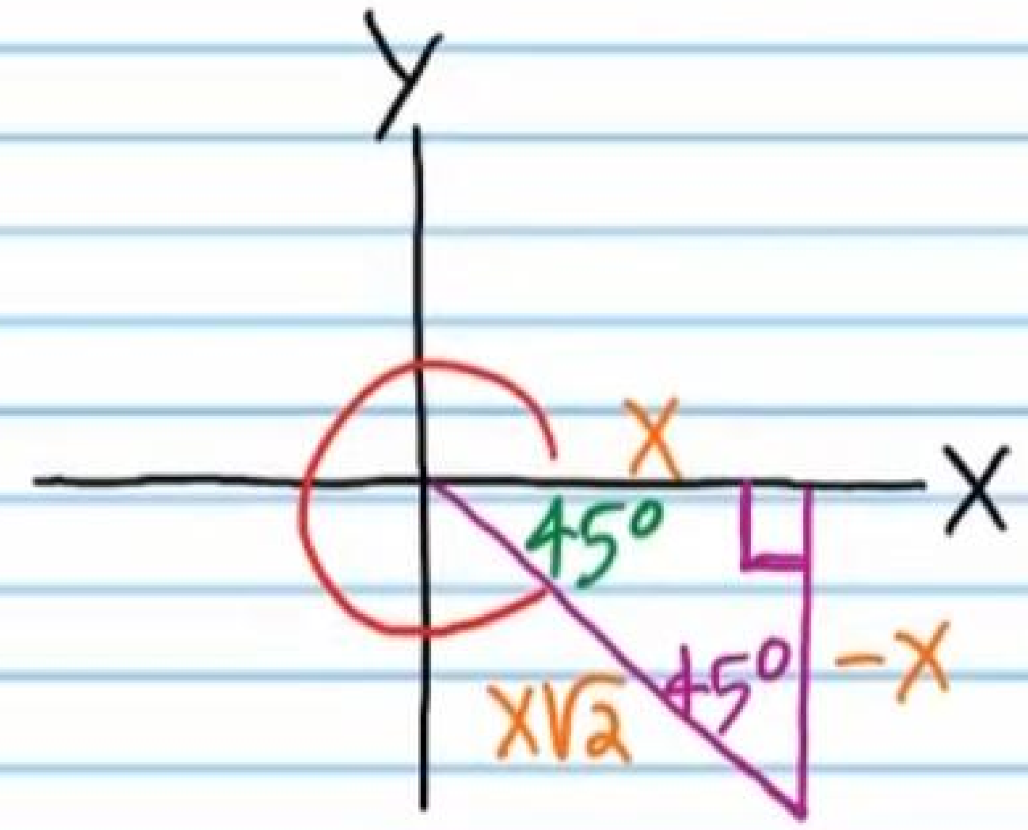
(28) If $\tan\theta = -1$, and $0° < \theta \leq 360°$, Find the Values of θ.



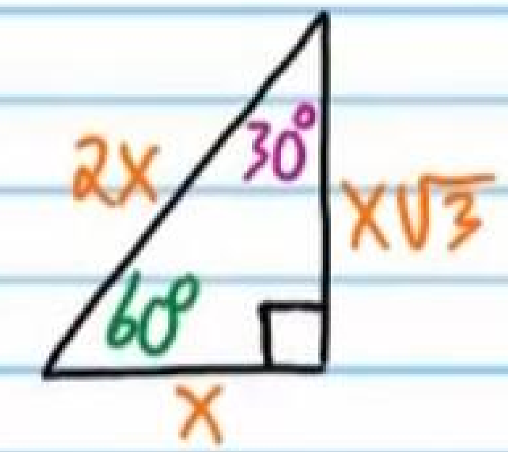


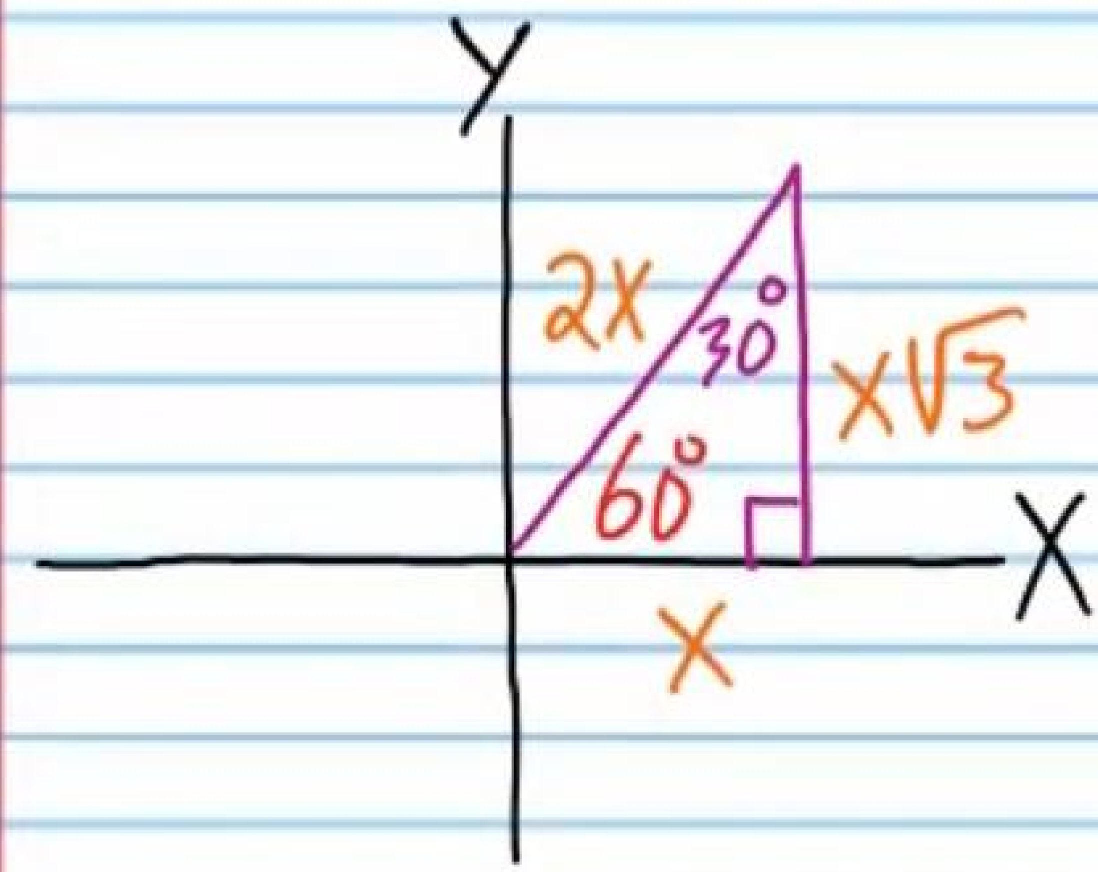


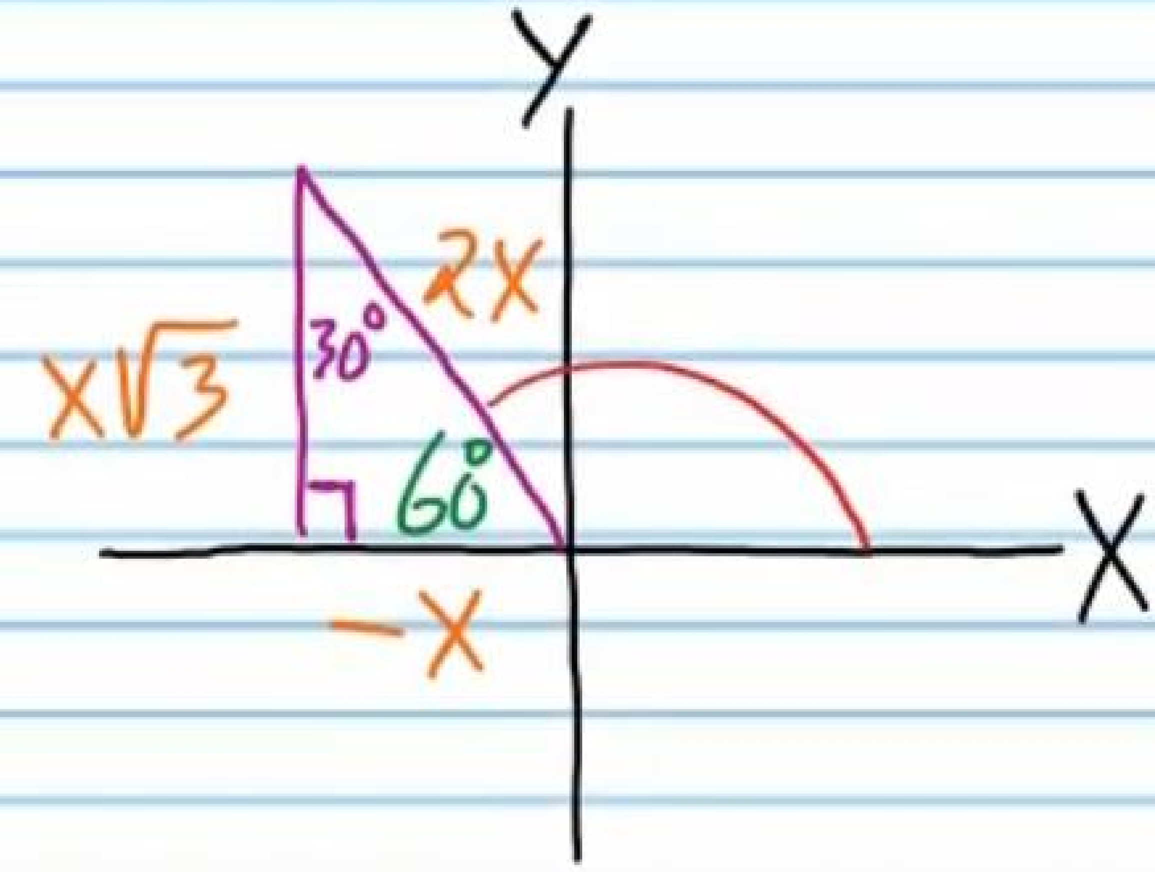


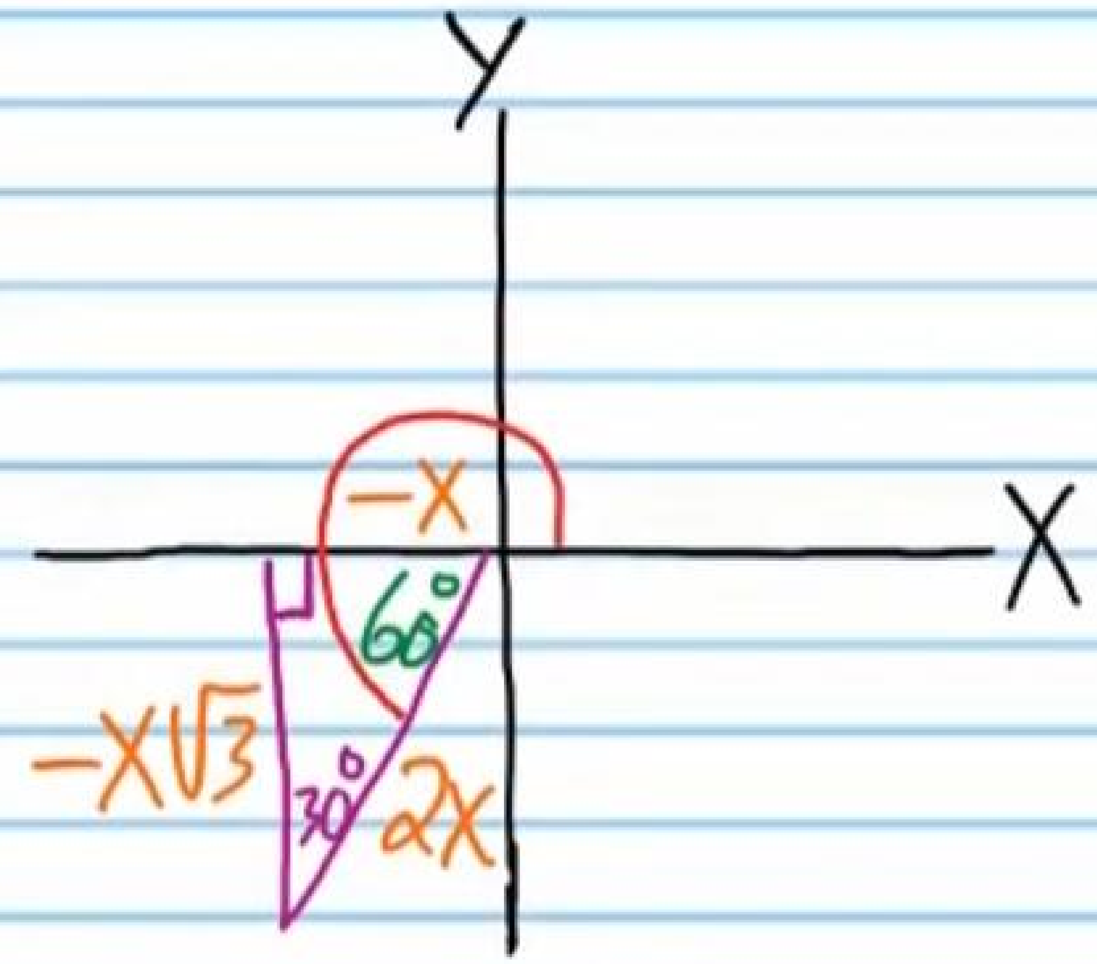


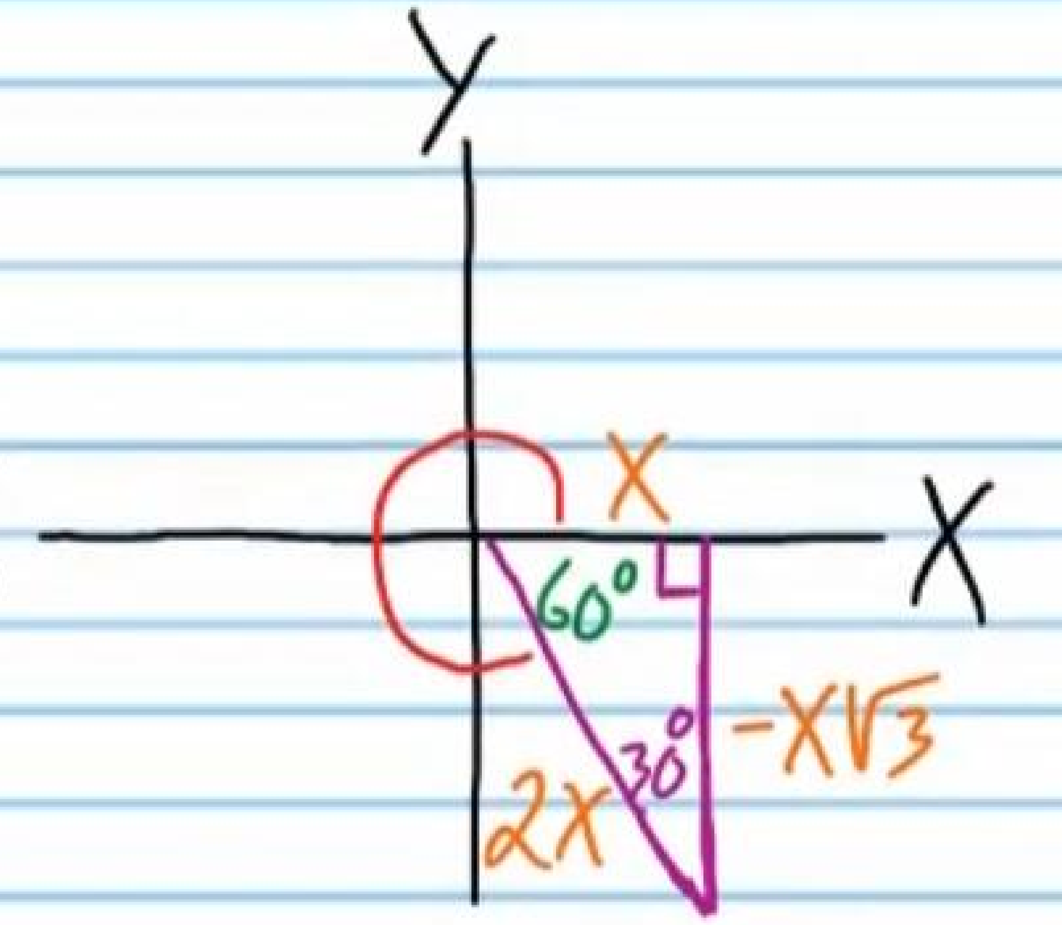
(29) If $\cot\theta = \frac{1}{\sqrt{3}}$, and $0° < \theta \leq 360°$, Find the Values of θ.



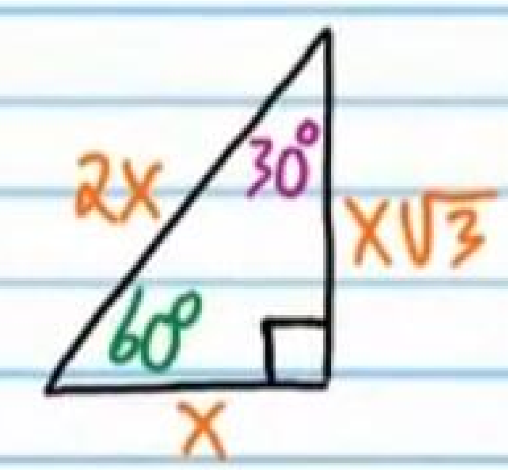


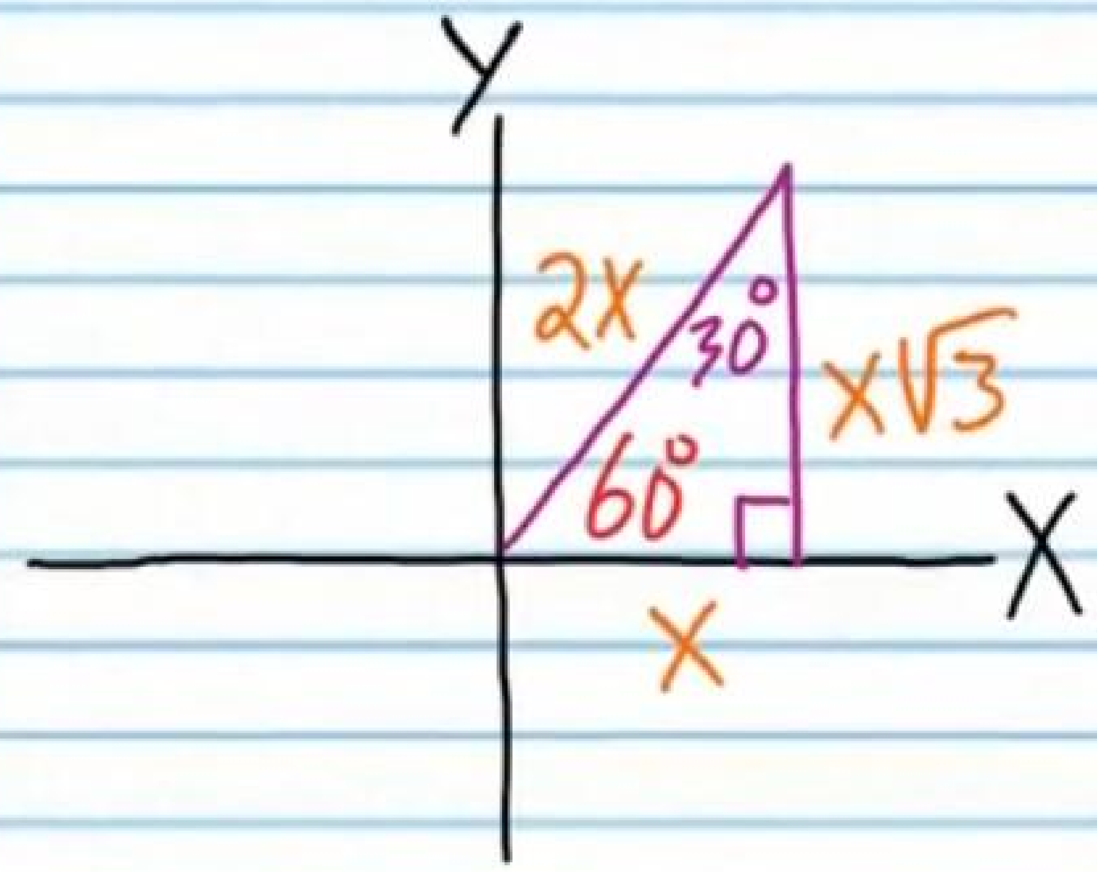


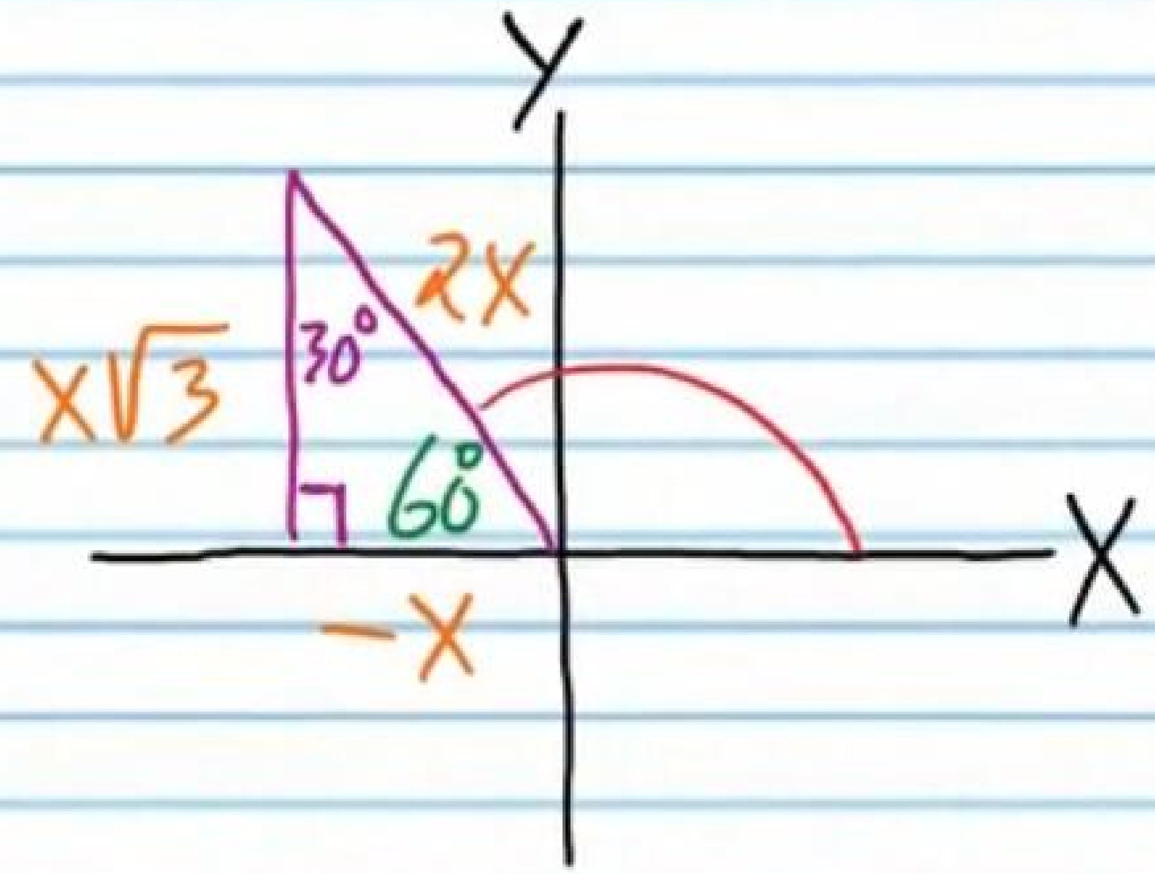


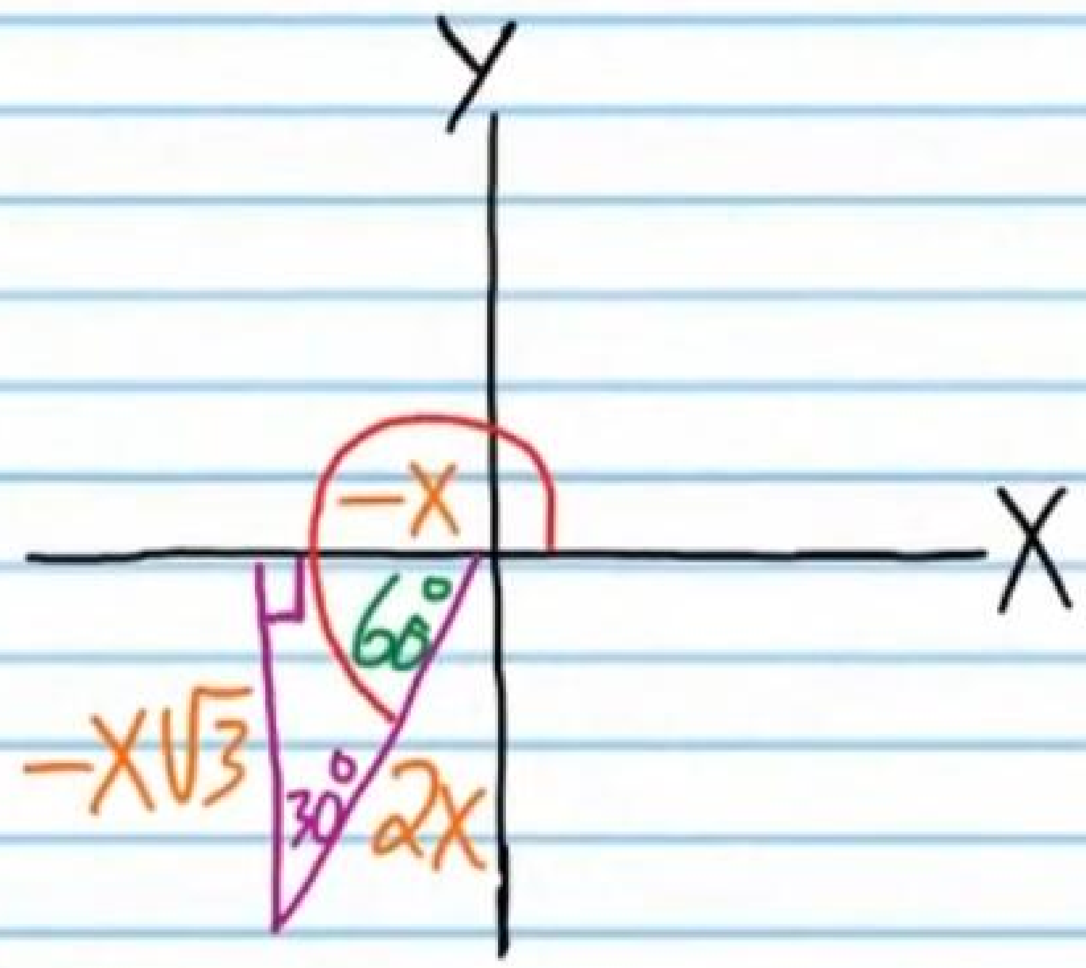


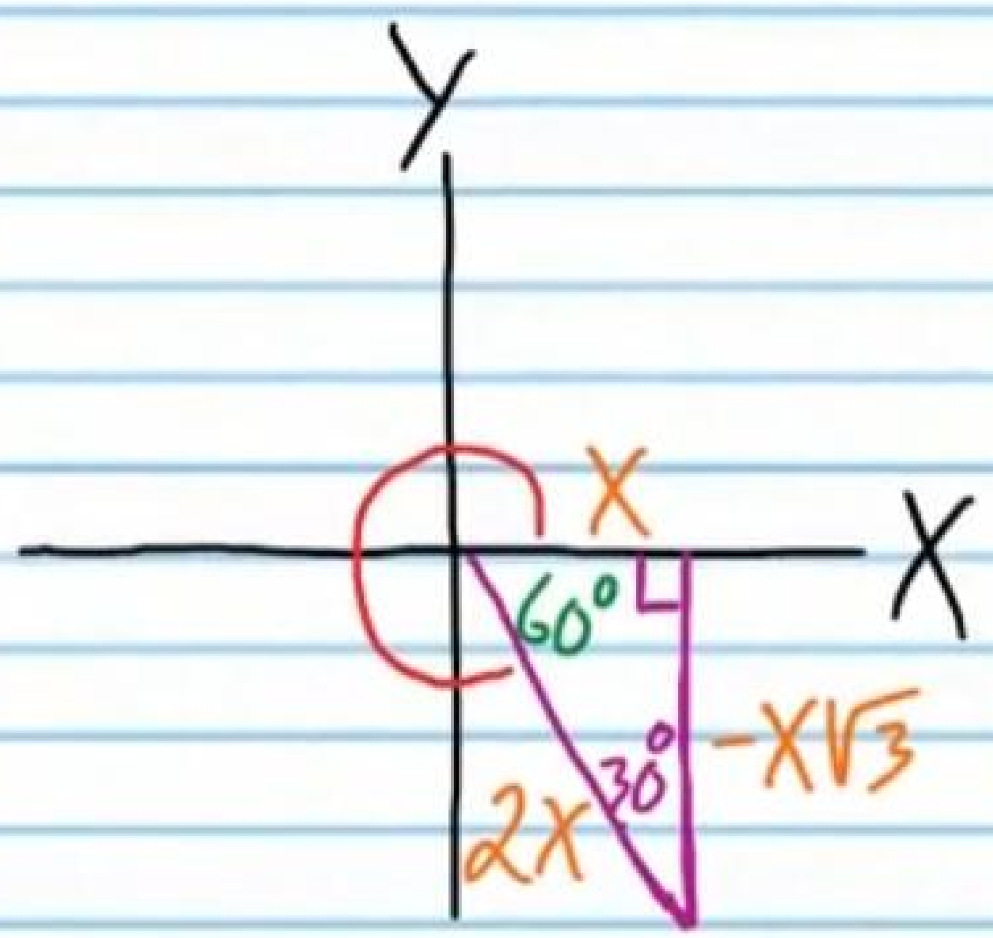
(30) If $\cos\theta = \frac{1}{2}$, and $0° < \theta \le 360°$, Find the Values of θ.





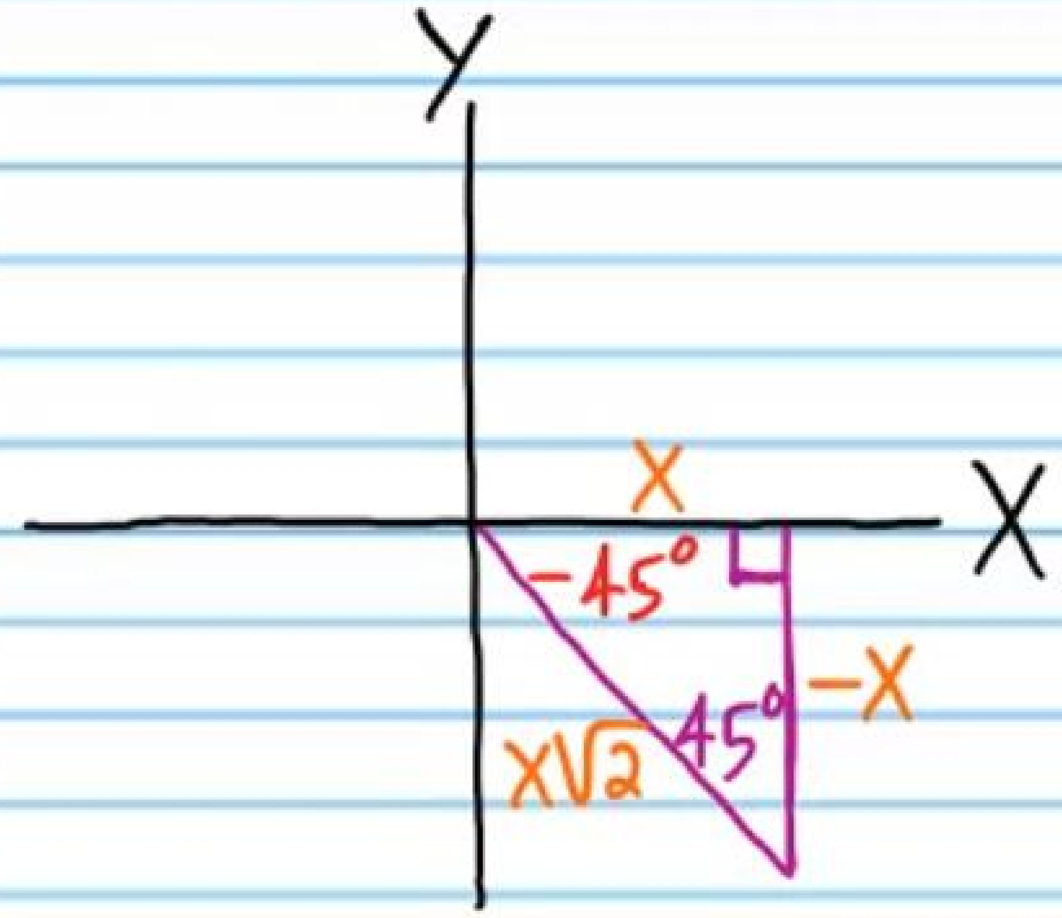


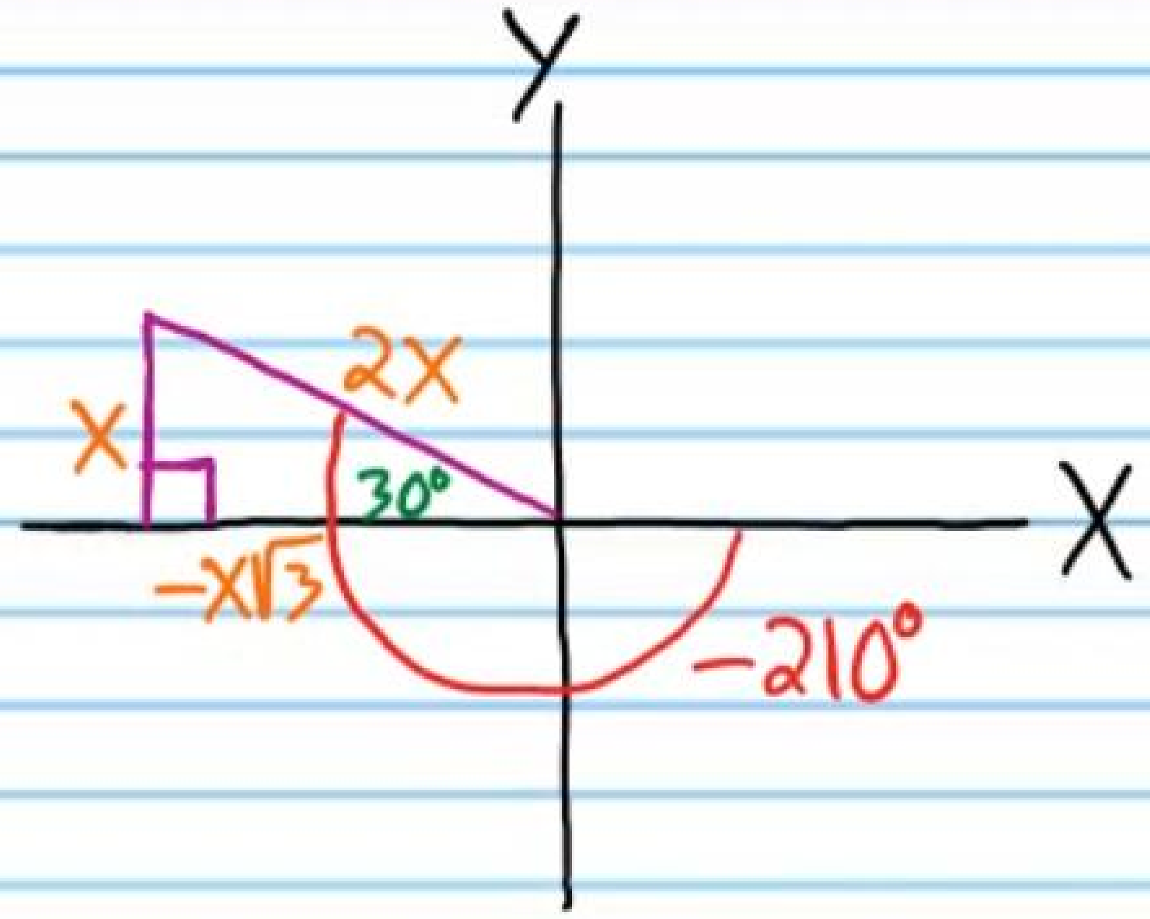




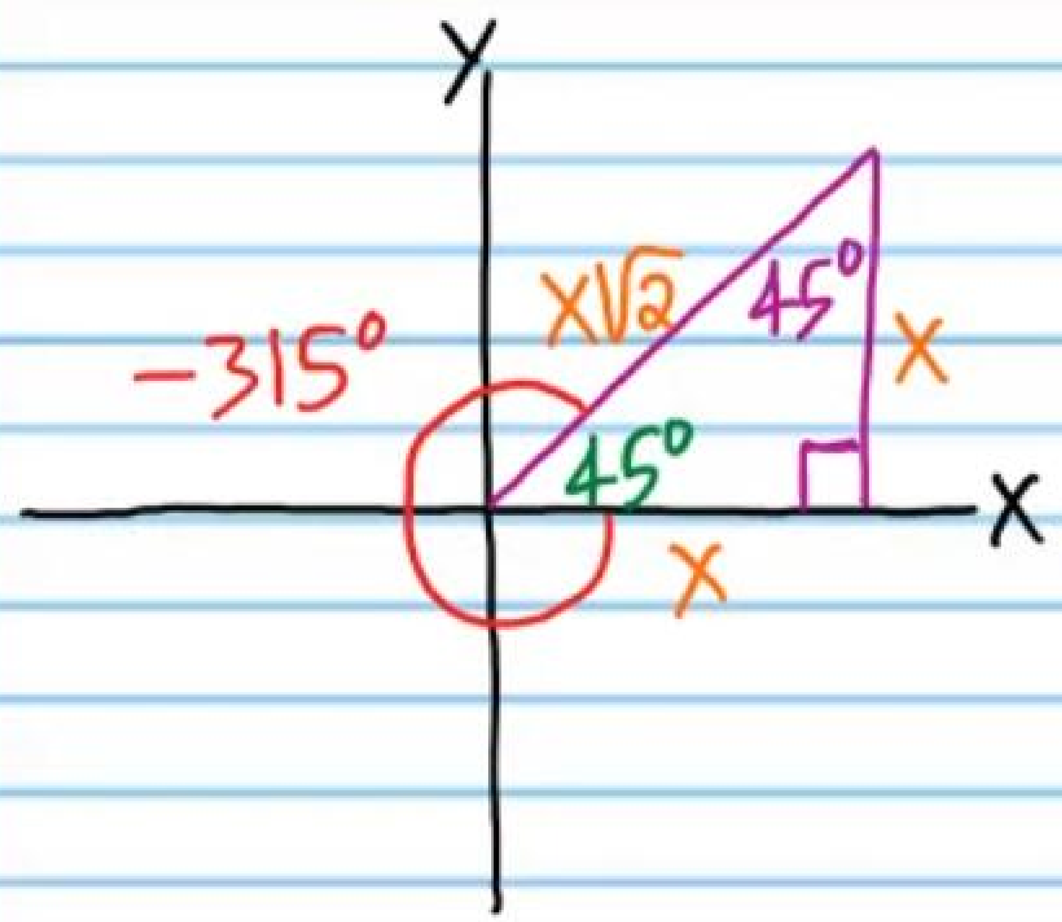
Homework Answers

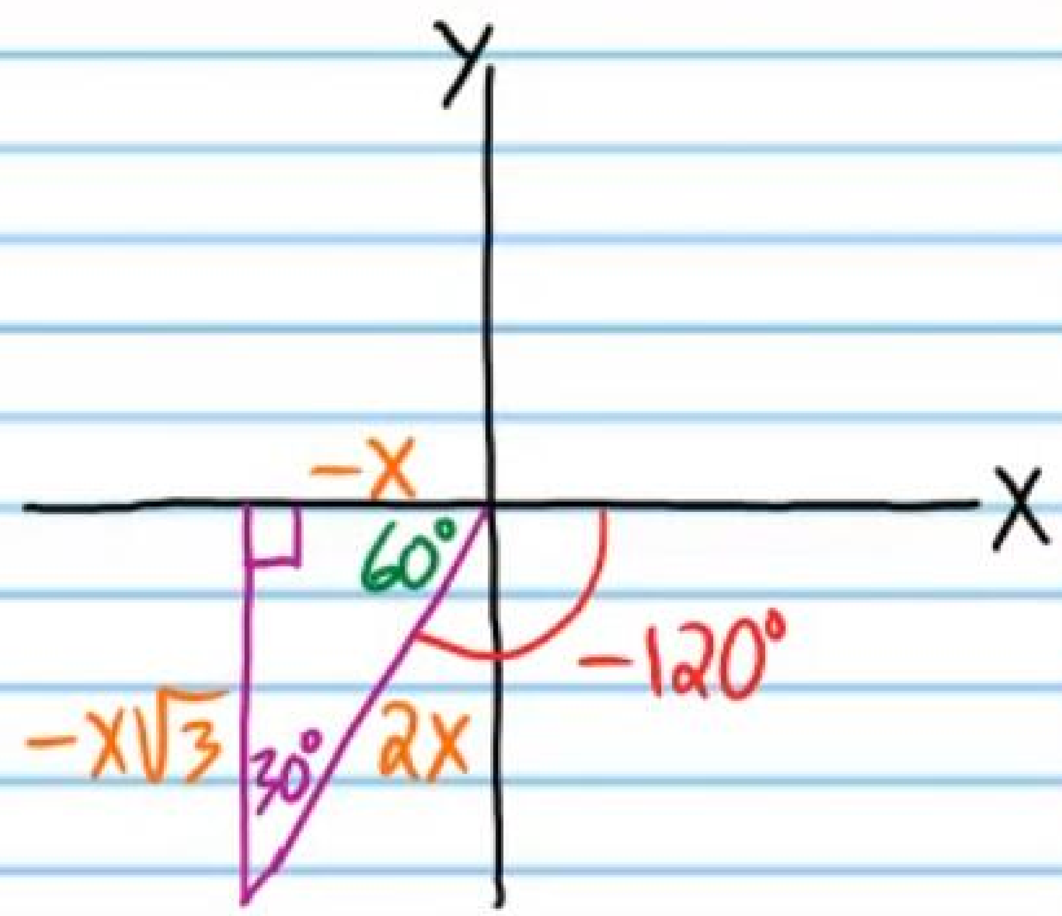
① $\cos(-45°) = \boxed{\frac{\sqrt{2}}{2}}$ ② $\cot(-210°) = \boxed{-\sqrt{3}}$





③ $\sin(-315°) = \boxed{\frac{\sqrt{2}}{2}}$ ④ $\sec(-120°) = \boxed{-2}$





⑤ $\boxed{1}$ ⑥ $\boxed{0}$ ⑦ $\boxed{0}$ ⑧ $\boxed{1}$ ⑨ $\boxed{0}$ ⑩ $\boxed{-1}$ ⑪ $\boxed{0}$

⑫ $\boxed{\text{Und}}$ ⑬ $\boxed{1}$ ⑭ $\boxed{\text{Und}}$ ⑮ $\boxed{1}$ ⑯ $\boxed{\text{Und}}$ ⑰ $\boxed{\text{Und}}$

⑱ $\boxed{\text{Und}}$ ⑲ $\boxed{0}$

Possible Answers (Need only three)

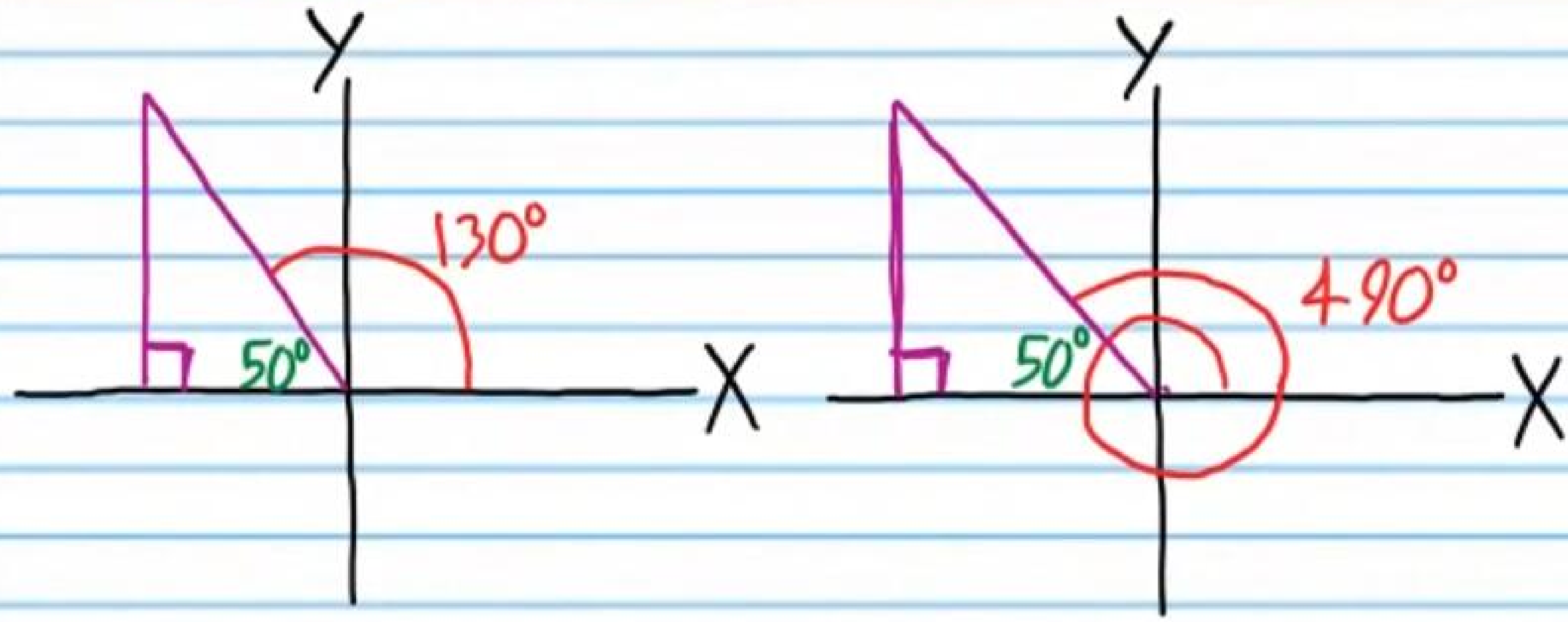
⑳ $\theta_1 = \boxed{-190°}$
$\theta_2 = \boxed{530°}$
$\theta_3 = \boxed{-550°}$
$\theta_4 = \boxed{890°}$

㉑ $\theta_1 = \boxed{318°}$
$\theta_2 = \boxed{678°}$
$\theta_3 = \boxed{-402°}$
$\theta_4 = \boxed{-762°}$

㉒ $\theta_1 = \boxed{-49°}$
$\theta_2 = \boxed{-409°}$
$\theta_3 = \boxed{671°}$
$\theta_4 = \boxed{1{,}031°}$
$\theta_5 = \boxed{-769°}$

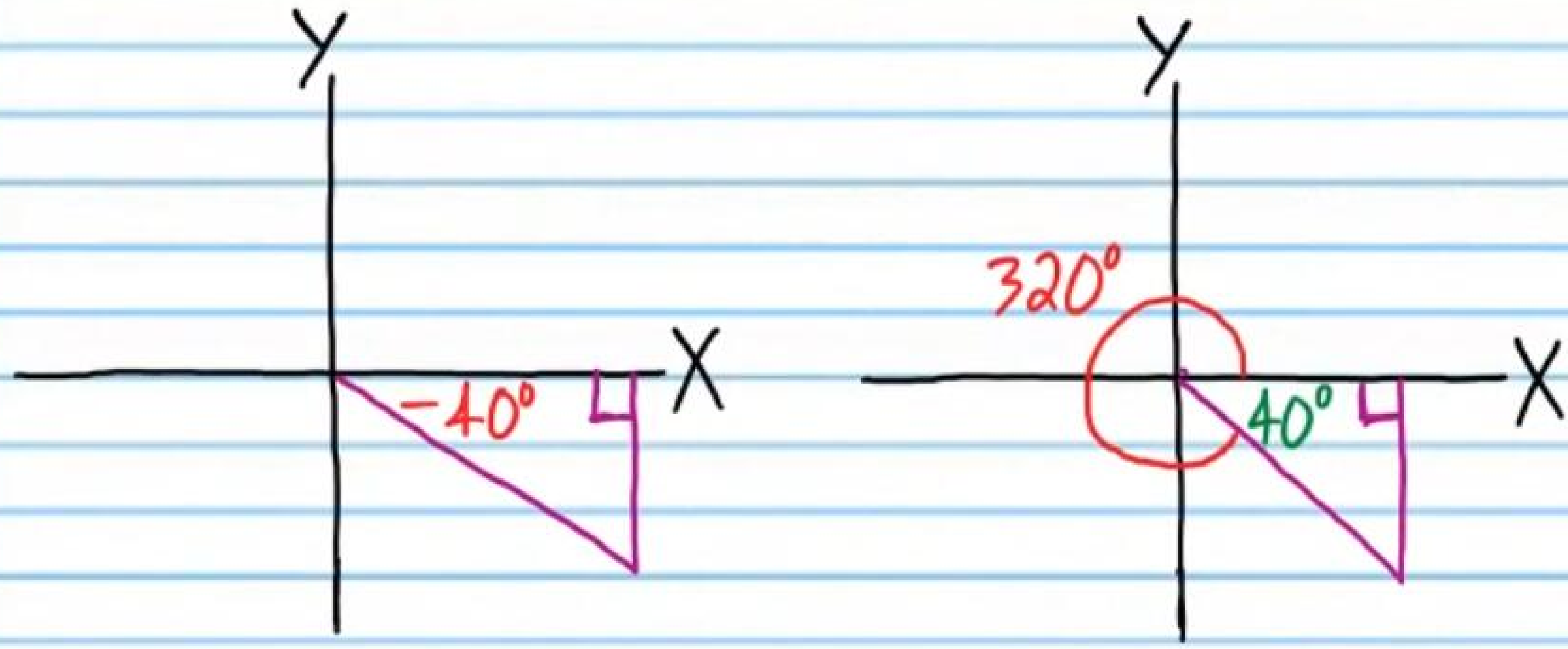
(23) $\sin 130° = \sin 490°$

$0.7660 = 0.7660$



(24) $\tan(-40°) = \tan 320°$

$-0.8391 = -0.8391$



(25) $\boxed{45°, 135°}$

(26) $\boxed{150°, 210°}$

(27) $\boxed{225°, 315°}$

(28) $\boxed{135°, 315°}$

(29) $\boxed{60°, 240°}$

(30) $\boxed{60°, 300°}$